جامعة الأخوين

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AL AKHAWAYN UNIVERSITY

School of Science and Engineering

CSC 5301: Advanced Database Systems & Data Warehousing

Business Intelligence System for Municipal Waste Management

Portfolio Edition

Prepared by:

Khawla Chrifi Alaoui

Developed as part of a collaborative academic project.

Under the supervision of:

Pr. Nasser Assem

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EXECUTIVE SUMMARY

The project delivers a comprehensive data-analytics model designed to support waste management operations across major Moroccan urban areas. The operational database captures structured city-level and regional data for key municipalities, including Casablanca, Rabat, Marrakech, Fes, Tangier, Agadir, Oujda, Kenitra, Tetouan, Safi, Beni Mellal, and Laayoune. This architecture enables the integration of raw operational data into a centralized analytical environment, transforming it into actionable insights that support evidence-based decision-making for municipal waste authorities and private waste collection companies.

Introduction

Urban waste management in Morocco faces increasing challenges due to rapid urbanization, population growth, and rising waste generation. Major cities such as Casablanca, Rabat, Marrakech, Fes, and Tangier require efficient, data-driven systems to monitor operations, control costs, and improve environmental sustainability. Traditional waste management approaches, which rely on fragmented and manual processes, limit the ability of municipalities to make informed decisions.

This project aims to design and implement an end-to-end business intelligence solution for municipal waste management using Microsoft SQL Server technologies. The solution integrates an operational database, a star-schema data warehouse, an SSIS-based ETL pipeline, an SSAS multidimensional cube, and reporting tools including SSRS and Power BI. Together, these components transform raw operational data into actionable insights for performance monitoring, cost analysis, and recycling assessment.

The remainder of this report presents the system architecture, data modeling, analytical processes, and reporting results, demonstrating how the proposed solution supports evidence-based decision-making in urban waste management.

Project Overview

1.2 Business Problem Statement

We were searching through a recent study to do my urban planning course, and it is evident that Moroccan urban centers are running plastic walls with garbage management. The combination of the growing population, an increase in trash production, and technological disparity is lethal to the system. The existing operations are in effect a shamble:

- Ineffective Route Planning: We do all the routes by hand; we are wasting fuel and money with no real purpose.
- Low Recycling Knowledge: We cannot trace in the streams of recyclable materials in the city hence the city is missing out on revenue.
- Low Service Monitoring: There exist no live measurements of the quality of service, so the individuals are not satisfied, and it is difficult to complain about the bad performance.
- Low-Cost Control: Under manual billing, the budget loses touch with reality.
- Poorest Data Integration: All the data in the operations are in silos, so there is difficulty in obtaining a complete picture or crunching the numbers.

1.3 Solution Architecture

The implemented solution follows a multi-tier architecture:

1. Operational Database (OLTP): An SQL Server database that records the data of transactions of the day-to-day activities.
2. Data Warehouse (DW): Star schema design for analytical processing
3. ETL Pipeline: SQL Server Integration Services (SSIS) for data transformation
4. Engine of analytics: SQL Server Analysis Services (SSAS) for multidimensional analysis
5. Reporting Layer: Power BI dashboards and SSRS reports for business intelligence.

2. PROJECT BACKGROUND AND CONTEXT

2.1 Moroccan Waste Management Landscape

2.1.1 Current Challenges

Morocco generates approximately 7 million tons of municipal solid waste annually, with urban areas producing 75% of this total. Key challenges include:

- Rapid Urbanization: 62% population living in urban areas with 2.1% annual growth rate.
- Waste Generation Increase: 3% annual increase in per capita waste generation.
- Limited Recycling Infrastructure: Only 10% recycling rate compared to European average of 46%.
- Service Coverage Gaps: 68% collection rate in urban areas, 22% in rural areas.
- Cost Pressures: Municipalities spend 10-15% of budgets on waste management.

2.1.2 Regulatory Environment

- National Waste Management Program: Target of 20% recycling rate by 2022, 30% by 2030.
- Law 28-00: Framework for waste management and environmental protection.
- Green Morocco Plan: Emphasis on sustainable development and circular economy.

2.2 Project Objectives

2.2.1 Primary Objectives

1. Operational Efficiency: Reduce collection costs by 15% through optimized routing.
2. Revenue Enhancement: Increase recycling revenue by 25% through better waste stream management.
3. Service Improvement: Achieve 95% service completion rate and reduce complaints by 30%.
4. Environmental Impact: Reduce carbon emissions by 20% through optimized operations.
5. Data Transparency: Provide real-time visibility into waste management operations.

3.2 Technology Stack Specification

3.2.1 Database Layer

- Database Engine: Microsoft SQL Server 2022 Enterprise Edition
- OLTP Database: WasteManagementOLTP (Normalized 3NF Design)
- Data Warehouse: WasteManagementDW (Star Schema Design)
- Backup Strategy: Daily full backups with 15-minute transaction log backups

3.2.2 ETL Layer

- ETL Tool: SQL Server Integration Services (SSIS) 2022
- Package Storage: SSIS Catalog (SSISDB)
- Execution Engine: SQL Server Agent for scheduled execution
- Performance: Parallel processing with batch operations

3.2.3 Analytics Layer

- OLAP Engine: SQL Server Analysis Services (SSAS) 2022 Multidimensional
- Cube Design: MOLAP storage for optimal query performance
- Processing: Incremental processing for daily updates
- Security: Role-based dimension security

3.2.4 Reporting Layer

- Dashboards: Power BI Desktop and Power BI Service
- Operational Reports: SQL Server Reporting Services (SSRS) 2022
- Scheduling: Report subscriptions and data-driven subscriptions

4. OPERATIONAL DATABASE (OLTP) DESIGN

4.1 OLTP Database Overview

The Operational Database (OLTP) is designed to support the daily transactional activities of municipal waste management operations. It captures detailed, real-time data related to waste collection processes, including routes, vehicles, employees, waste types, collection events, and operational incidents.

The OLTP system is implemented using Microsoft SQL Server and follows a normalized relational model (Third Normal Form – 3NF) to ensure data consistency, reduce redundancy, and maintain transactional integrity. This database serves as the primary source of truth for

operational data and acts as the foundation for downstream analytical processing in the Data Warehouse.

4.2 Entity-Relationship Model

The Entity-Relationship (ER) model represents the logical structure of the OLTP database by defining entities, attributes, and relationships between them. The model is centered around waste collection operations and connects geographic entities (cities, zones), operational entities (routes, trucks, employees), and event-based entities (collection events, fuel usage, incidents).

Relationships are enforced using primary and foreign keys to ensure referential integrity. The ER model supports one-to-many and many-to-one relationships, reflecting real-world operational dependencies such as multiple collection events occurring on a single route or a truck being assigned to many collection operations over time.

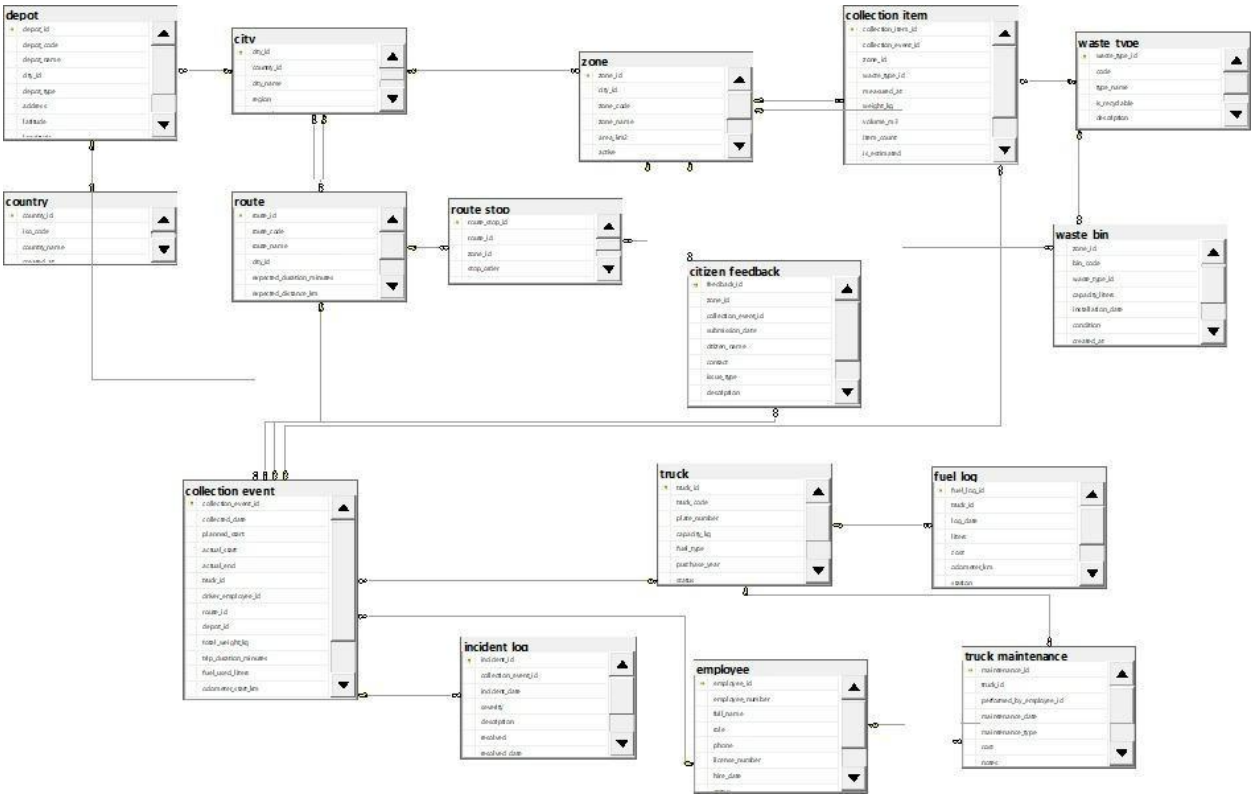


Figure 1: Entity Relationship Diagram

4.3 Core OLTP Tables Description

4.3.1 City Table

Stores information about cities where waste management operations are conducted. It defines the highest geographic level used for reporting and operational planning.

4.3.2 Zone Table

Represents administrative zones within a city. Each zone belongs to a single city and is used for operational routing, performance monitoring, and service coverage analysis.

4.3.3 Depot Table

Contains information about waste depots or collection centers. Depots act as operational hubs where trucks are dispatched and waste is delivered.

4.3.4 Route Table

Defines predefined waste collection routes. Routes are associated with zones and depots and represent planned paths for collection activities.

4.3.5 Route Stop Table

Stores detailed stop-level information for each route, including the sequence of stops. This table enables fine-grained route analysis and optimization.

4.3.6 Employee Table

Maintains records of employees involved in waste collection, primarily drivers and operational staff. It includes identification and employment-related attributes.

4.3.7 Truck Table

Stores details of collection vehicles, including capacity, fuel type, and operational status. Trucks are assigned to routes and collection events.

4.3.8 Waste Type Table

Defines categories of waste, such as organic, plastic, paper, or mixed waste. It includes recyclability indicators to support recycling analysis.

4.3.9 Collection Event Table

Represents a single waste collection operation performed by a truck and driver on a specific route and date. It serves as a central transactional entity.

4.3.10 Collection Item Table

Stores item-level waste collection data associated with a collection event. This allows tracking multiple waste types and quantities per event.

4.3.11 Waste Bin Table

Contains information about waste bins distributed across zones, including bin type and capacity. It supports monitoring of infrastructure coverage.

4.3.12 Fuel Log Table

Records fuel consumption details for trucks during collection operations. This table is essential for cost analysis and fuel efficiency monitoring.

4.3.13 Incident Log Table

Captures operational incidents such as missed collections, vehicle breakdowns, or delays. It supports service quality evaluation.

4.3.14 Truck Maintenance Table

Stores maintenance and servicing records for trucks. It helps track vehicle reliability and maintenance costs.

4.3.15 Citizen Feedback Table

Contains complaints and feedback submitted by citizens regarding waste collection services. This table supports service performance evaluation and accountability.

Data Visualization

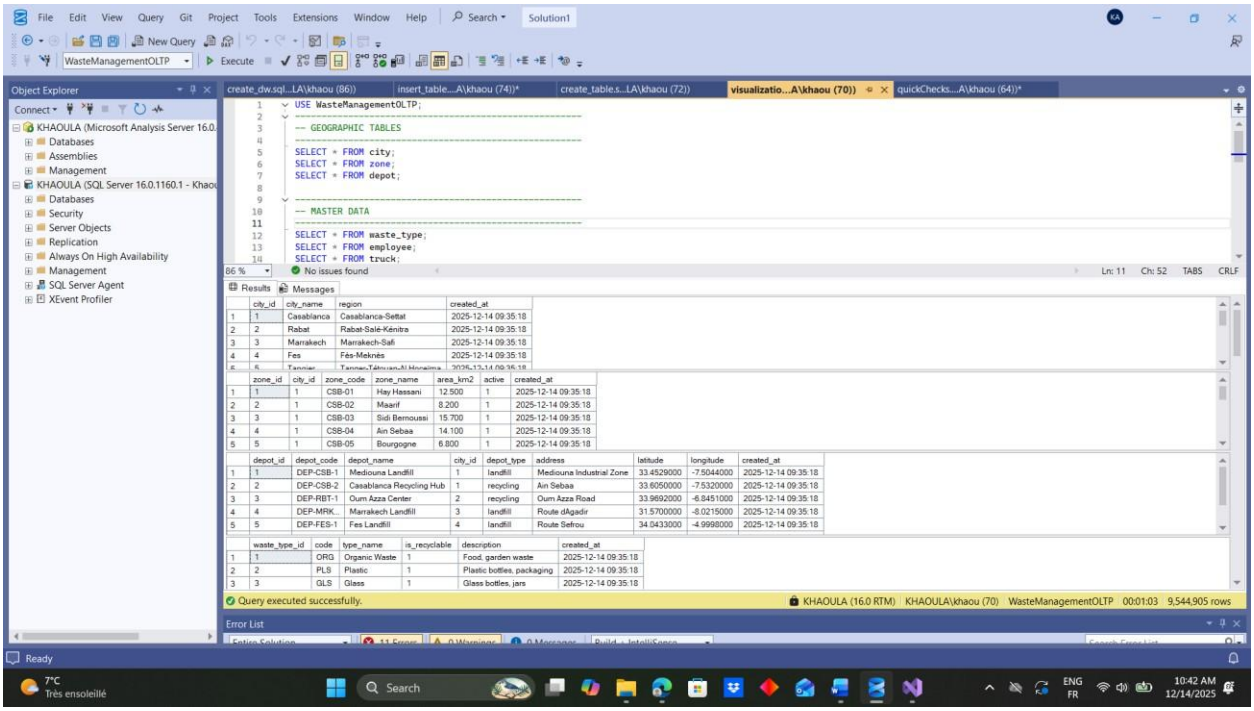


Figure 2: Tables Visualization (city, zone, and depot)

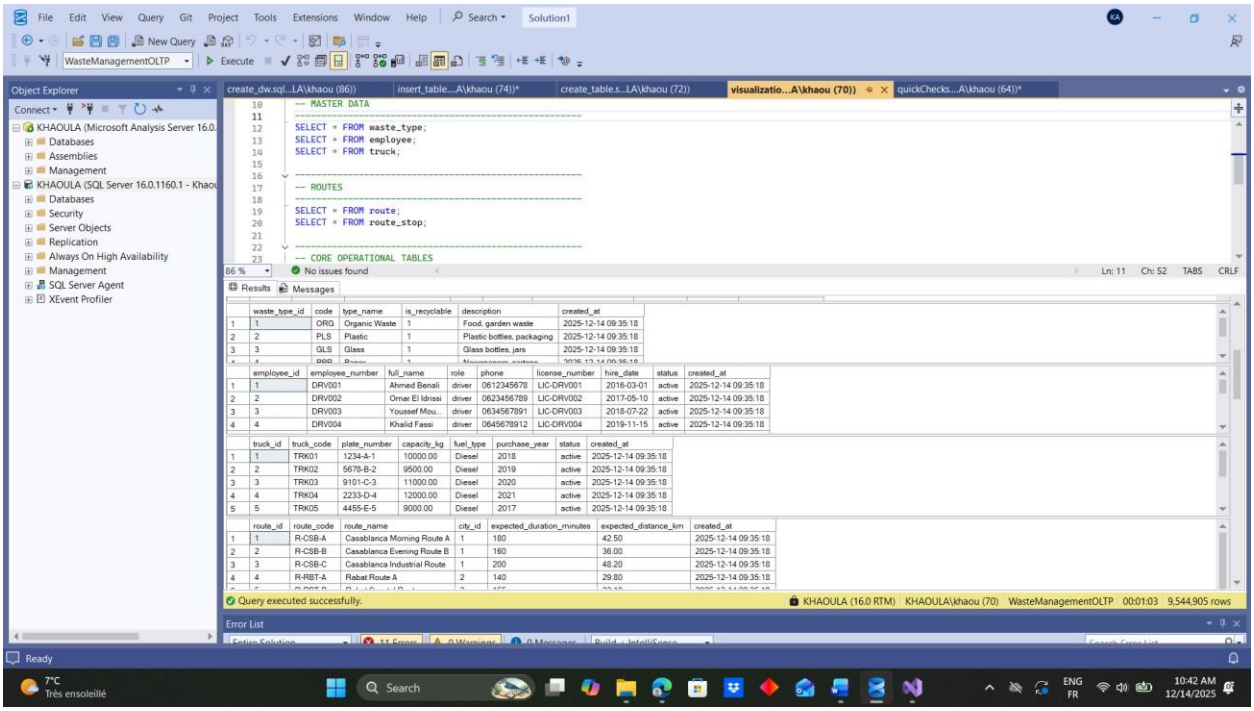


Figure 3: Tables Visualization (waste_type, employee, and truck)

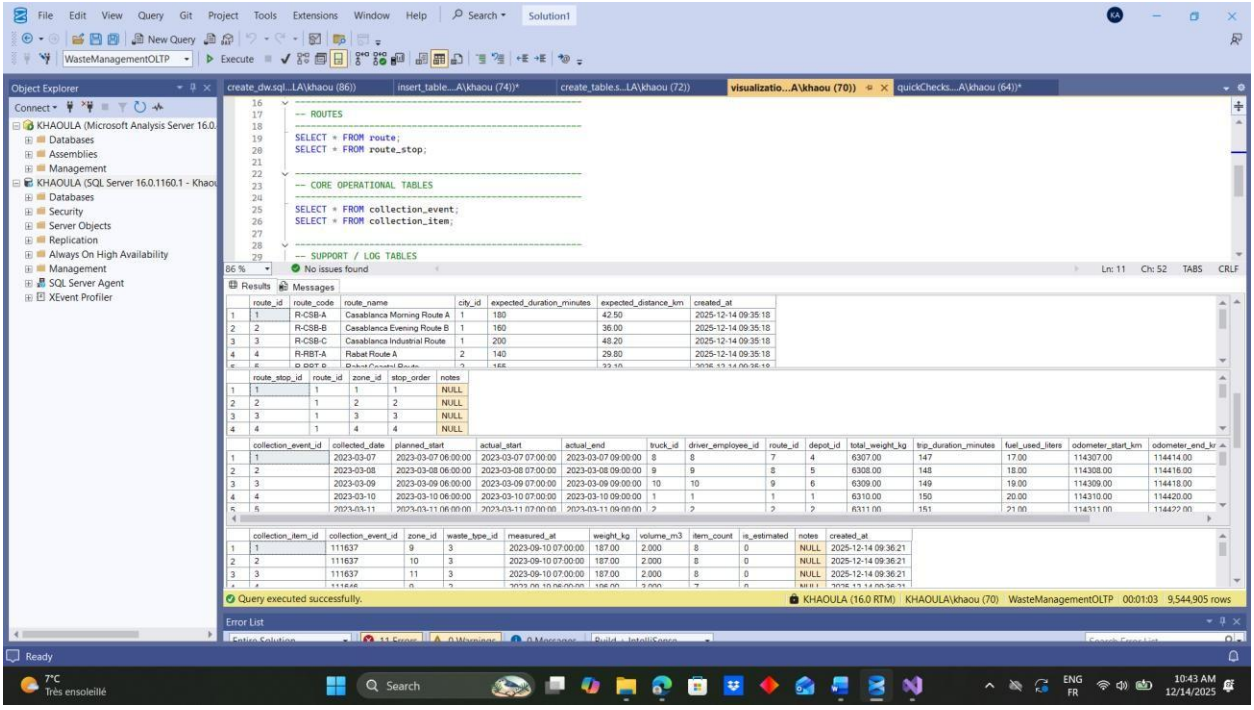


Figure 4: Tables Visualization (route, route_stop, collection_event, and collection_item)

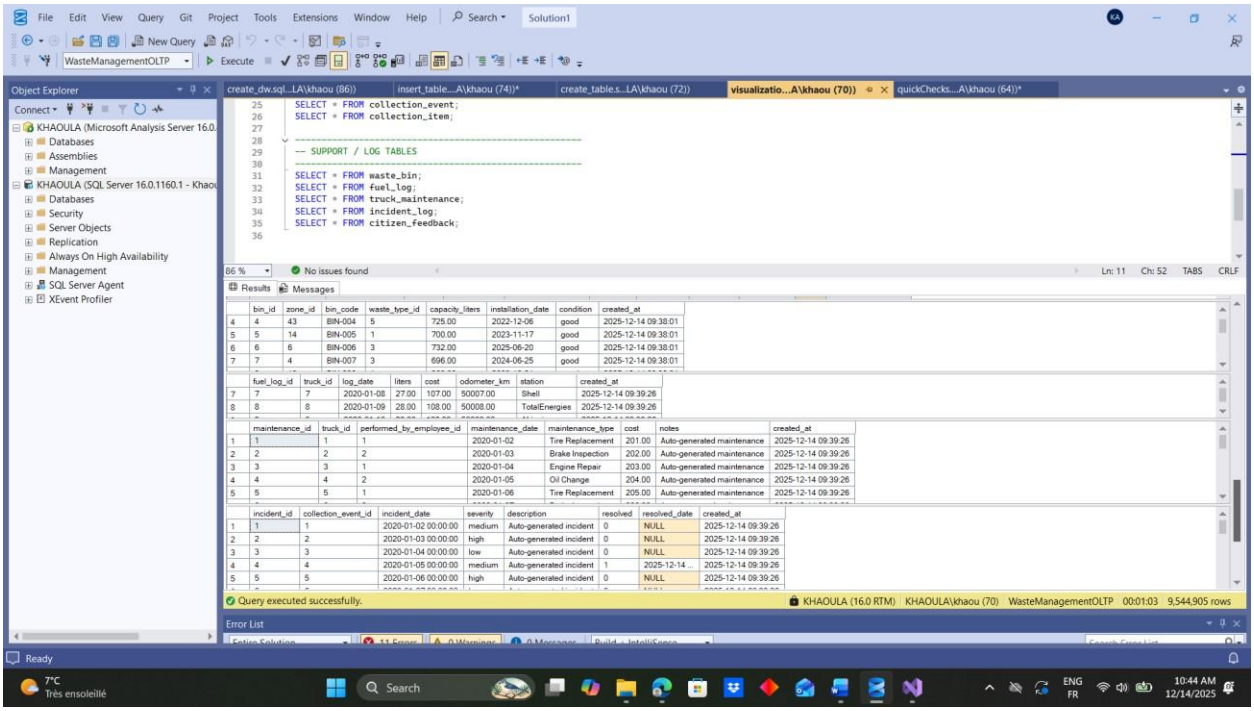


Figure 5: Tables Visualization (waste_bin, fuel_log, truck_maintenance, incident_log, and citizen_feedback)

4.4 Primary and Foreign Key Relationships

The OLTP database enforces data consistency and referential integrity through a well-defined system of primary keys (PKs) and foreign keys (FKs). These constraints ensure that all operational records are valid, traceable, and correctly linked across entities involved in waste management activities.

Each table is uniquely identified by a primary key, while foreign keys establish logical relationships between dependent entities. This structure accurately models real-world operational dependencies and prevents data anomalies such as orphan records or inconsistent references.

4.4.1 Geographic Relationships

- The City table uses city_id as its primary key.
- The Zone table references the City table through city_id, establishing a one-to-many relationship, where one city can contain multiple zones.
- This relationship ensures that every zone belongs to a valid city and supports geographic aggregation of operational data.

4.4.2 Infrastructure and Routing Relationships

- The Depot table is identified by depot_id and represents operational hubs.
- The Route table references both zone_id and depot_id, linking each route to a specific geographic zone and a depot.
- The Route_Stop table references route_id, allowing multiple stops to be associated with a single route in a defined sequence.

These relationships model the physical flow of waste collection activities and enable route-level performance analysis.

4.4.3 Human Resources and Vehicle Relationships

- The Employee table is identified by employee_id.
- The Truck table is identified by truck_id.
- Both entities are referenced by the Collection_Event table, which links a specific driver and truck to each operational event.

This design allows tracking driver performance, vehicle utilization, and assignment history.

4.4.4 Operational Event Relationships

- The Collection_Event table serves as a central transactional entity and is identified by collection_event_id.
- It references:
    - route_id → Route
    - truck_id → Truck
    - employee_id → Employee
- The Collection_Item table references collection_event_id and waste_type_id, enabling multiple waste types to be recorded within a single collection event.

This structure supports detailed operational tracking while maintaining normalization.

4.4.5 Waste and Resource Tracking Relationships

- The Waste_Type table is identified by waste_type_id and categorizes collected waste.
- The Waste_Bin table references zone_id, linking physical bins to geographic zones.
- The Fuel_Log table references truck_id, associating fuel consumption records with specific vehicles.

- The Truck_Maintenance table also references truck_id, ensuring that all maintenance activities are tied to valid trucks.

These relationships enable cost control, resource optimization, and sustainability analysis.

4.4.6 Service Quality and Feedback Relationships

- The Incident_Log table references collection_event_id, allowing incidents to be traced back to specific operations.
- The Citizen_Feedback table references zone_id, enabling spatial analysis of complaints and service quality issues.

This ensures that operational issues and citizen satisfaction can be analyzed in a structured and accountable manner.

4.5 Data Integrity and Normalization Strategy

The Operational Database (OLTP) of the Waste Management Information System was designed following strict data integrity and normalization principles to ensure accuracy, consistency, and reliability of operational data prior to analytical processing. Given that this database serves as the primary source for the data warehouse, maintaining high data quality at this level is essential.

4.5.1 Normalization Strategy

The OLTP database follows the principles of Third Normal Form (3NF) to minimize data redundancy and prevent update, insertion, and deletion anomalies. Each entity represents a single business concept and stores only attributes that are functionally dependent on the primary key.

- First Normal Form (1NF)

  All tables contain atomic attributes, and repeating groups are eliminated. For example, waste collection details are separated into *Collection Event* and *Collection Item* tables rather than stored in a single denormalized structure.

- Second Normal Form (2NF)

  Non-key attributes fully depend on the entire primary key. Composite relationships, such as route stops and collection items, are decomposed into separate tables to avoid partial dependencies.

- Third Normal Form (3NF)

  Transitive dependencies are removed by isolating reference data (e.g., cities, zones, waste types, trucks, employees) into independent lookup tables. This ensures that descriptive attributes are not duplicated across multiple tables.

This normalized structure allows efficient transactional operations and ensures clean, consistent source data for downstream ETL processes.

4.5.2 Data Integrity Enforcement

To guarantee data correctness and consistency, multiple integrity constraints are implemented at the database level:

- Primary Key Constraints

  Each table is defined with a primary key to uniquely identify records and prevent duplicate entries.

- Foreign Key Constraints

  Referential integrity is enforced through foreign keys that define valid relationships between tables. For example:

  - Collection events reference valid trucks, drivers, routes, and depots.
  - Collection items reference valid waste types, zones, and collection events.
  - Operational logs such as fuel logs and maintenance records reference existing trucks.

- Mandatory and Optional Attributes

  NOT NULL constraints are applied to critical operational fields (e.g., identifiers, dates, quantities) to ensure that essential information is always captured, while optional attributes are allowed where business rules permit flexibility.

4.5.3 Consistency and Domain Control

To maintain logical consistency across the system:

- Controlled Reference Tables are used for entities such as waste types, cities, zones, and depots, ensuring standardized values throughout the database.

- Data Type Enforcement ensures that numerical, date, and textual fields store values in appropriate formats, preventing invalid or inconsistent data entries.
- Logical Separation of Processes ensures that operational data (transactions) is clearly separated from descriptive and reference data.

4.5.4 Readiness for Analytical Processing

By applying normalization and integrity constraints at the OLTP level, the database provides a reliable and consistent source for the ETL layer. This design significantly reduces the complexity of data cleansing during extraction and transformation, enabling efficient loading into the star-schema-based data warehouse.

Overall, the OLTP normalization and integrity strategy ensures data accuracy, reduces redundancy, and establishes a strong foundation for scalable analytics and business intelligence.

5. DATA WAREHOUSE DESIGN

5.1 Data Warehouse Overview

The Data Warehouse (DW) is designed to support analytical querying, reporting, and decision-making for waste management operations. Unlike the OLTP system, which focuses on transaction processing, the DW is optimized for read-heavy workloads, historical analysis, and performance measurement.

The warehouse follows a dimensional modeling approach, enabling fast aggregations, simplified queries, and compatibility with OLAP tools such as SSAS, SSRS, and Power BI.

5.2 Star Schema Architecture

The DW adopts a star schema architecture, consisting of a central fact table surrounded by multiple dimension tables. This design simplifies analytical queries by reducing joins and improves query performance.

At the center lies the Fact_Waste_Operations table, which captures quantitative metrics of waste collection activities. Dimension tables provide descriptive context such as time, location, driver, truck, and waste type.

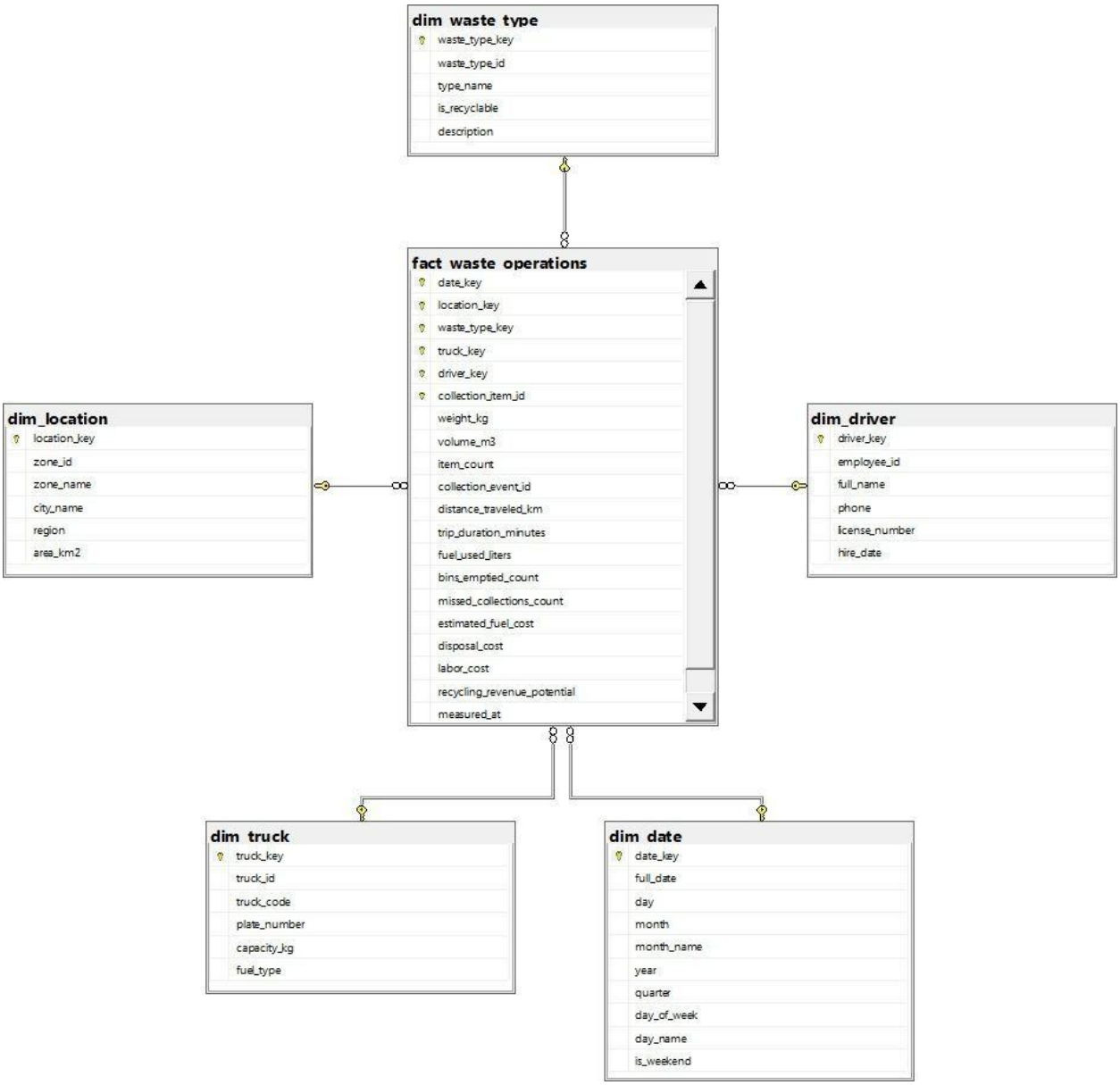


Figure 6: Star Schema

5.3 Fact Table Design

5.3.1 Fact_Waste_Operations Table

The Fact_Waste_Operations table is the central fact table of the Data Warehouse and captures all quantitative measures related to waste collection operations. It integrates operational data originating from multiple OLTP entities such as collection events, collection items, trucks, drivers, routes, and depots.

The table contains the following categories of measures:

- Waste metrics: collected weight (weight_kg), volume (volume_m3), and item count

- Operational metrics: trip duration, distance traveled, fuel consumption
- Cost metrics: estimated fuel cost, disposal cost, and labor cost
- Environmental indicators: recycling revenue potential
- Service indicators: bins emptied and missed collections

Each record is linked to all analytical dimensions through surrogate keys, ensuring fast aggregations and multidimensional analysis.

5.3.2 Fact Table Granularity Definition:

The granularity of the fact table is defined as:

One waste collection item collected at a specific date, in a specific zone, by a specific driver and truck, for a given waste type

This fine-grained level of detail allows:

- Accurate operational analysis
- Flexible aggregation (daily, monthly, city-level, driver-level)
- Advanced KPIs such as efficiency and fuel productivity

5.4 Dimension Tables Design

5.4.1 Date Dimension

The Dim_Date table provides a complete calendar structure and is referenced by all fact records using date_key (YYYYMMDD format).
It includes attributes such as day, month, month name, year, quarter, day of week, and weekend indicators.

This dimension enables:

- Time-series analysis
- Trend identification
- Seasonal performance evaluation

5.4.2 Location Dimension

The Dim_Location table consolidates geographical data by combining zone, city, and region attributes.
It also includes zone surface area, allowing density and productivity analysis.

This dimension supports:

- City and zone performance comparison
- Regional waste generation analysis
- Spatial efficiency assessment

5.4.3 Driver Dimension

The Dim_Driver table stores descriptive information about collection drivers, including full name, phone number, license details, and hire date.

It enables:

- Workforce performance evaluation
- Driver efficiency analysis
- Accountability and operational monitoring

5.4.4 Truck Dimension

The Dim_Truck table describes the vehicle fleet used for waste collection. Attributes include truck code, plate number, capacity, and fuel type.

This dimension supports:

- Fleet utilization analysis
- Fuel efficiency comparison
- Maintenance and capacity planning

5.4.5 Waste Type Dimension

The Dim_Waste_Type table categorizes collected waste and indicates recyclability. It is essential for environmental impact analysis and sustainability reporting.

This dimension enables:

- Recycling performance measurement
- Waste composition analysis
- Revenue potential estimation

5.5 Surrogate Keys

All dimension tables use surrogate keys generated within the Data Warehouse to decouple analytical data from OLTP identifiers.

This ensures historical consistency and improves performance.

6. DATA INTEGRATION USING SQL SERVER INTEGRATION SERVICES (SSIS)

6.1 ETL Architecture Overview

SSIS is used as the ETL engine to extract data from the OLTP system, transform it according to business rules, and load it into the Data Warehouse.

The ETL process is modular, scalable, and scheduled using SQL Server Agent.

6.2 Data Extraction from OLTP System

Data is extracted from normalized OLTP tables such as:

- Collection events
- Collection items
- Employees
- Trucks
- Locations
- Waste types

Optimized SQL queries ensure minimal impact on transactional operations.

6.3 Data Transformation Logic

Transformation steps include:

- Data type conversions
- Lookup transformations to resolve surrogate keys
- Derived column calculations (KPIs, costs)
- Data standardization and cleansing

6.4 Data Loading into the Data Warehouse

Dimensions are loaded first, followed by the fact table to maintain referential integrity.

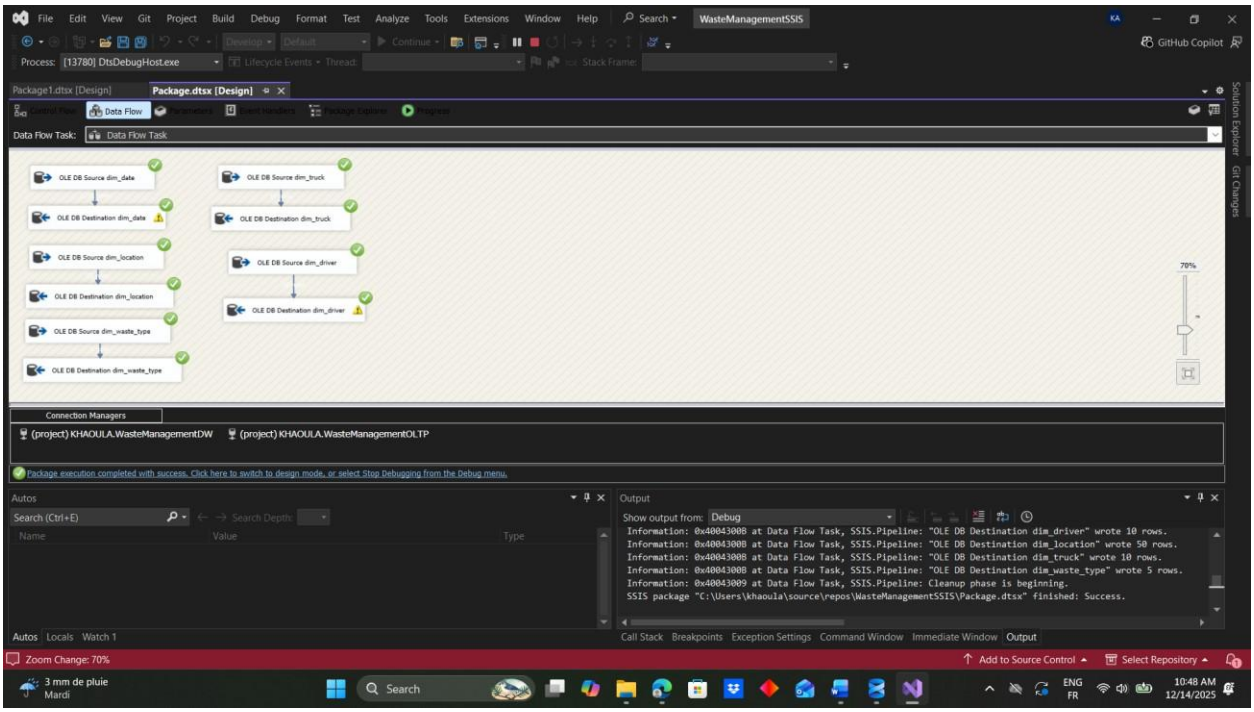


Figure 7: Dimensions Loading

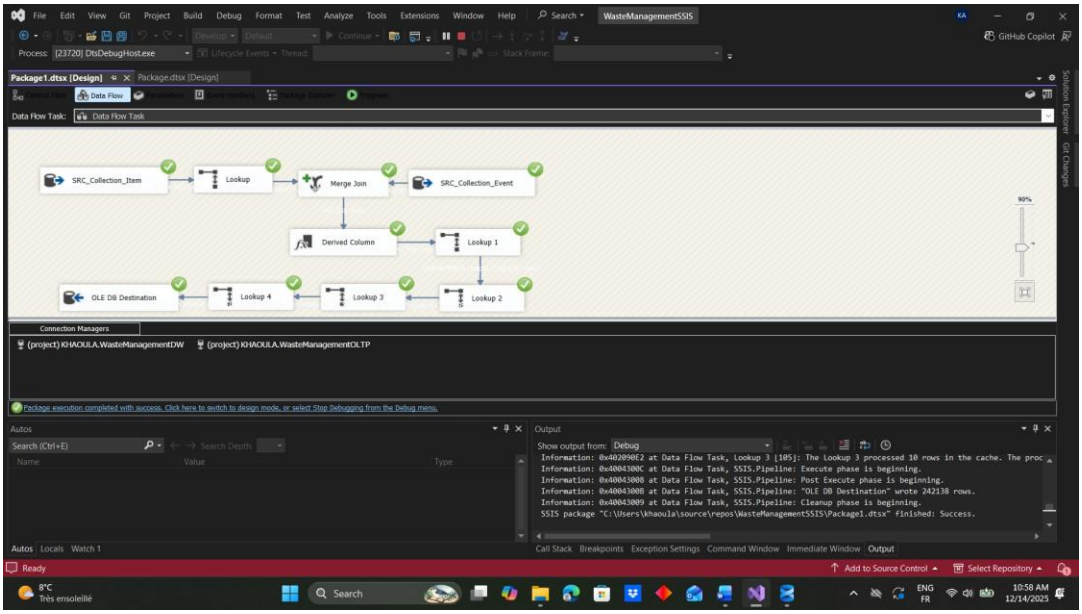


Figure 8: Fact Table Loading

Explanation

Lookup (Waste Type Lookup)

The first Lookup transformation is used to enrich the incoming collection item data with attributes from the dim_waste_type dimension table in the data warehouse.

The lookup matches the operational field waste_type_id from the source data with the corresponding waste_type_id in the dimension table. As a result, two additional attributes are retrieved:

- waste_type_key (the surrogate key used in the fact table)
- is_recyclable (a descriptive attribute used later for analytics)

This step ensures that operational identifiers from the OLTP system are correctly translated into surrogate keys required by the star schema design. Records that do not find a match can be redirected or handled separately to preserve referential integrity.

Merge Join (Collection Item + Collection Event)

The Merge Join transformation combines data from two different OLTP sources:

- SRC_Collection_Item (item-level waste measurements)
- SRC_Collection_Event (trip-level operational details)

An inner join is performed using the common field collection_event_id, ensuring that only valid and complete records are propagated downstream.

This transformation consolidates granular waste measurements (weight, volume, item count) with contextual trip information such as:

- truck identifier
- driver identifier
- trip duration
- fuel consumption
- route and depot information

The merge join establishes the complete operational context required to populate the fact_waste_operations table.

Derived Column Transformation

The Derived Column transformation is used to compute additional analytical measures that do not exist directly in the OLTP system.

Based on the configuration:

- estimated_fuel_cost is calculated using fuel consumption
- disposal_cost is derived from collected waste weight
- labor_cost is calculated from trip duration
- recycling_revenue_potential is conditionally calculated based on recyclability
- measured_date is extracted from the original timestamp

These derived metrics represent business rules and cost models and are intentionally calculated during ETL to centralize logic, ensure consistency, and reduce computational load during reporting and analysis.

Lookup 1 (Date Dimension Lookup)

Lookup 1 resolves the relationship between the operational date (measured_date) and the Date Dimension (dim_date).

The transformation matches the operational date to the corresponding full_date in the date dimension and retrieves the associated date_key.

This step enables:

- time-based aggregation (daily, monthly, yearly)
- trend analysis
- alignment with SSAS time hierarchies

The resulting date_key is stored in the fact table as a foreign key.

Lookup 2 (Location Dimension Lookup)

Lookup 2 maps the operational zone_id to the Location Dimension (dim_location).

Using the lookup, the transformation retrieves:

- location_key (surrogate key)
- associated descriptive attributes such as city name and region (available in the dimension)

This ensures that each fact record is correctly associated with a geographical context, enabling city-, zone-, and region-level analysis in reports and dashboards.

Lookup 3 (Truck Dimension Lookup)

Lookup 3 enriches the dataset with the surrogate key from the Truck Dimension (dim_truck).

The operational truck_id is matched against the dimension table, and the corresponding truck_key is returned. This enables:

- truck performance analysis
- fuel efficiency evaluation
- maintenance and capacity-based reporting

The lookup decouples operational identifiers from analytical keys, reinforcing the dimensional model.

Lookup 4 (Driver Dimension Lookup)

The final Lookup 4 resolves driver information by matching the operational driver_employee_id with the Driver Dimension (dim_driver).

The transformation retrieves the driver_key, which allows the fact table to reference driver records using surrogate keys. This enables:

- driver productivity analysis
- workload distribution assessment
- performance benchmarking across drivers

This lookup completes the dimensional enrichment process before loading data into the fact table.

6.5 ETL Automation

To ensure that analytical data remains up to date, an automated processing mechanism was configured using SQL Server Agent. A scheduled job named WasteManagement_Autorefresh was created to periodically refresh the SQL Server Analysis Services (SSAS) cube.

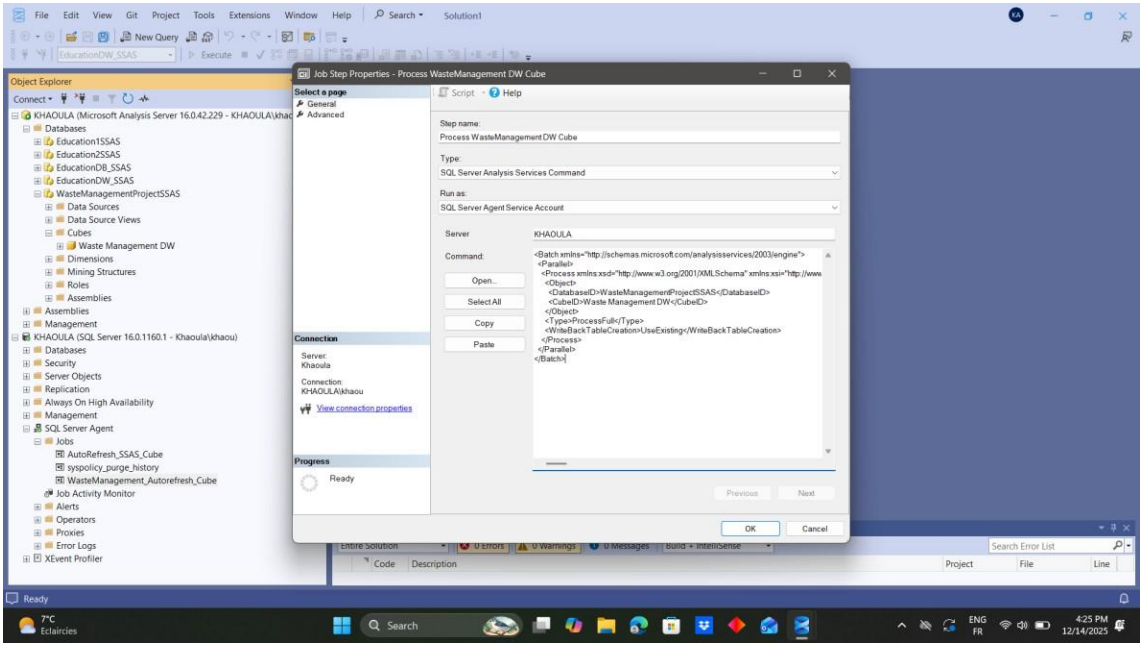


Figure 9: Job's Setup

The *WasteManagement_Autorefresh* SQL Server Agent job is configured to run once every day at 1:00 AM.

This schedule ensures the SSAS cube is refreshed regularly with the latest available data while avoiding repeated executions within the same day. Running the job once daily provides a balance between data freshness and system performance, and it is suitable when source data is updated daily.

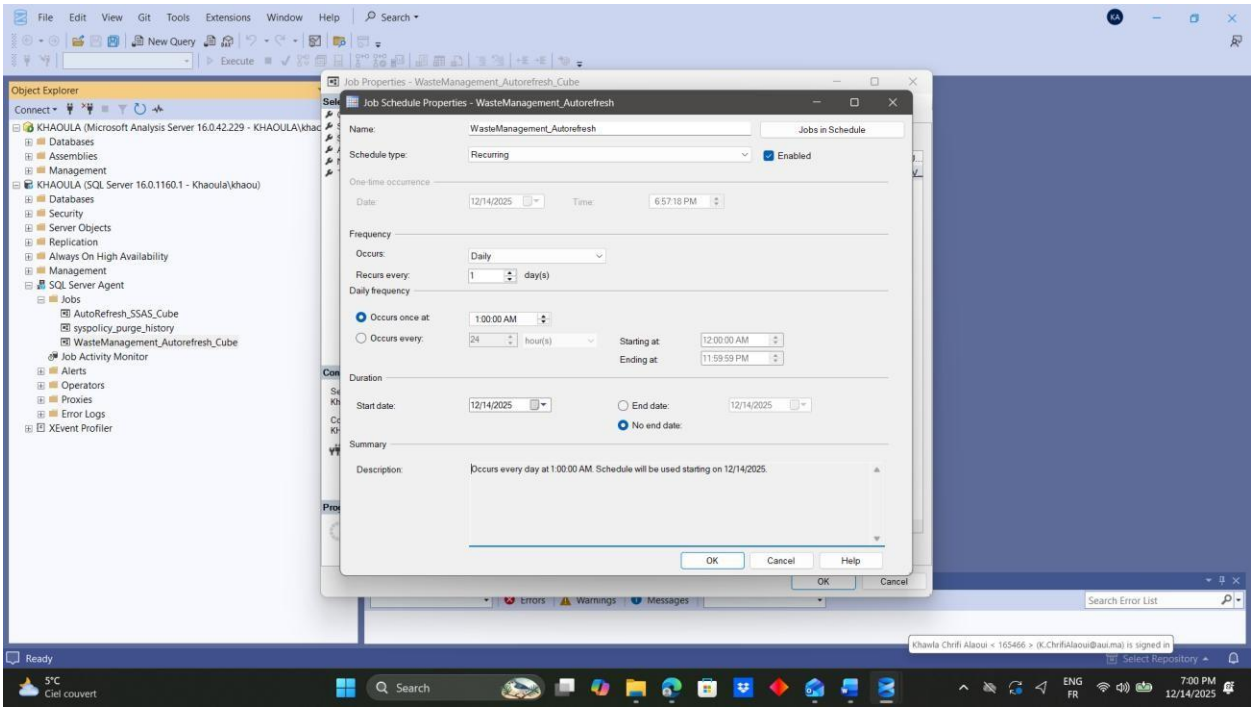


Figure 10: Daily SSAS Cube Auto-Refresh Scheduled at 1:00 AM

Finally, *WasteManagement_Autorefresh* SQL Server Agent job was manually executed to validate its configuration. The job completed successfully without errors, confirming that the SSAS cube processing task is correctly configured and operational. This manual execution ensures the job is ready for automated daily execution as scheduled.

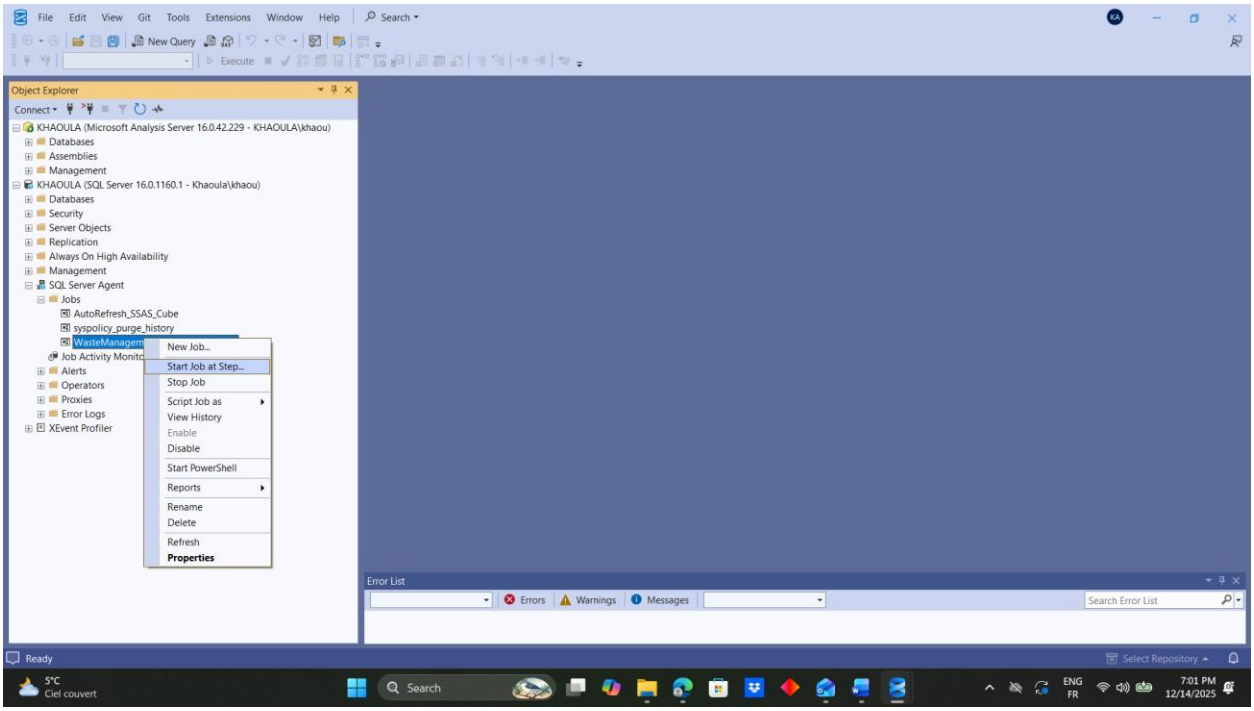


Figure 11: Manual Start of SQL Server Agent Job for SSAS Cube Auto-Refresh

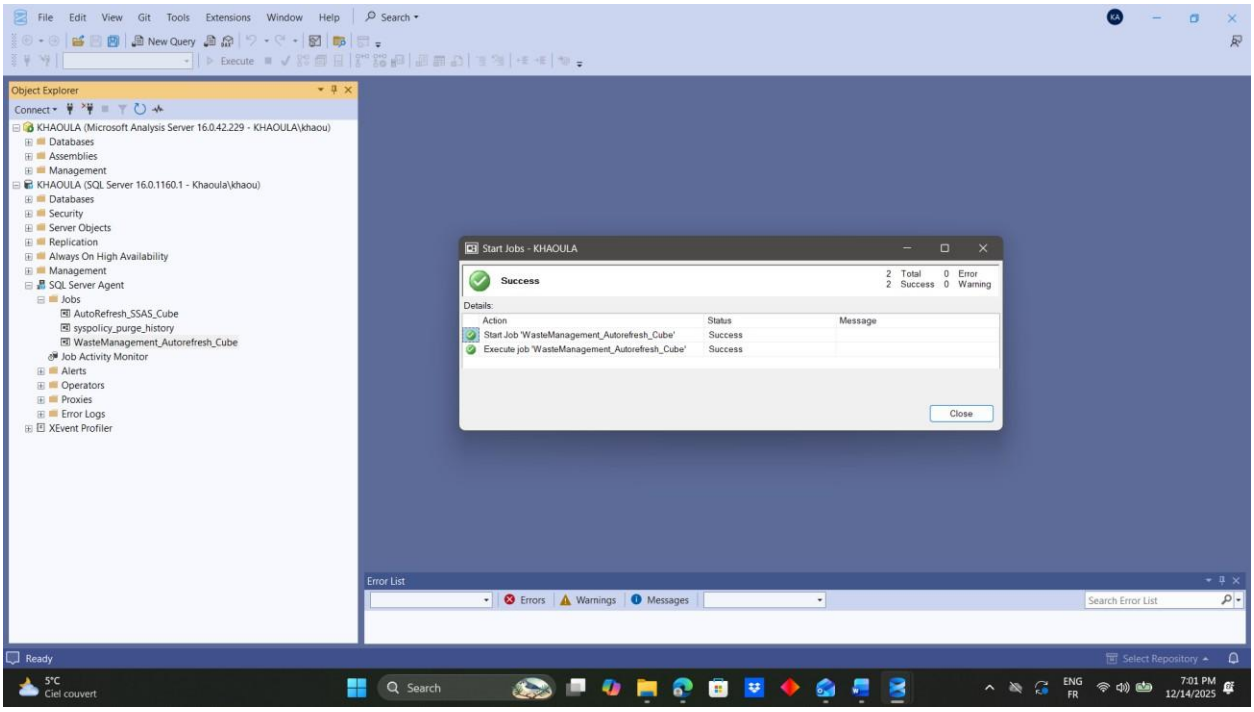


Figure 12: Successful Completion of the Manual Execution of the SSAS Auto-Refresh Job

7. ANALYTICS USING SQL SERVER ANALYSIS SERVICES (SSAS)

7.1 Multidimensional Cube Architecture

A multidimensional cube is built on top of the Data Warehouse using SSAS.

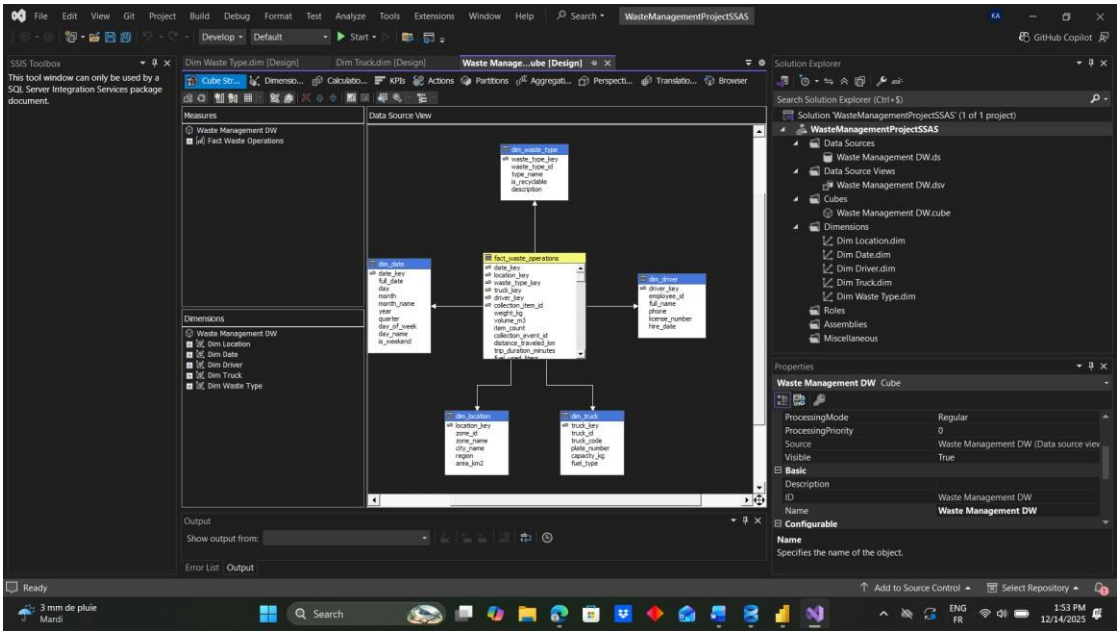


Figure 13: Waste Management DW Cube

We have successfully deployed the project and processed the cube

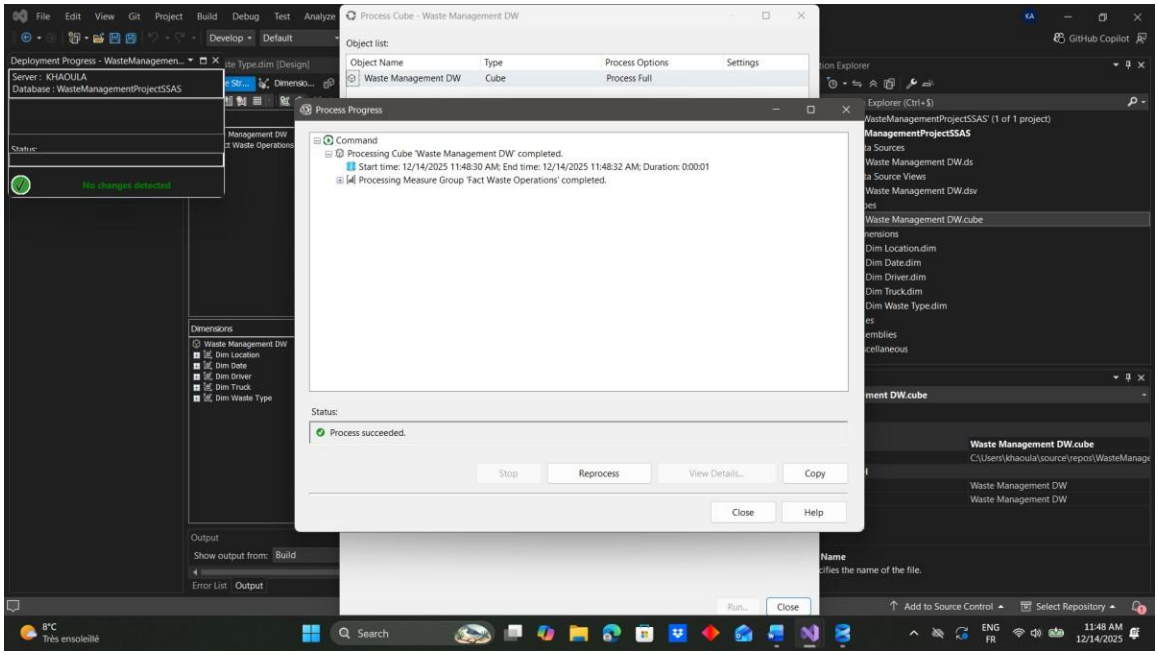


Figure 14: Project's Deployment & Cube's Processing Success

The screenshot below confirms that data was successfully transferred from our source database “WasteManagementOLTP” to our target data warehouse “WasteManagementDW”.

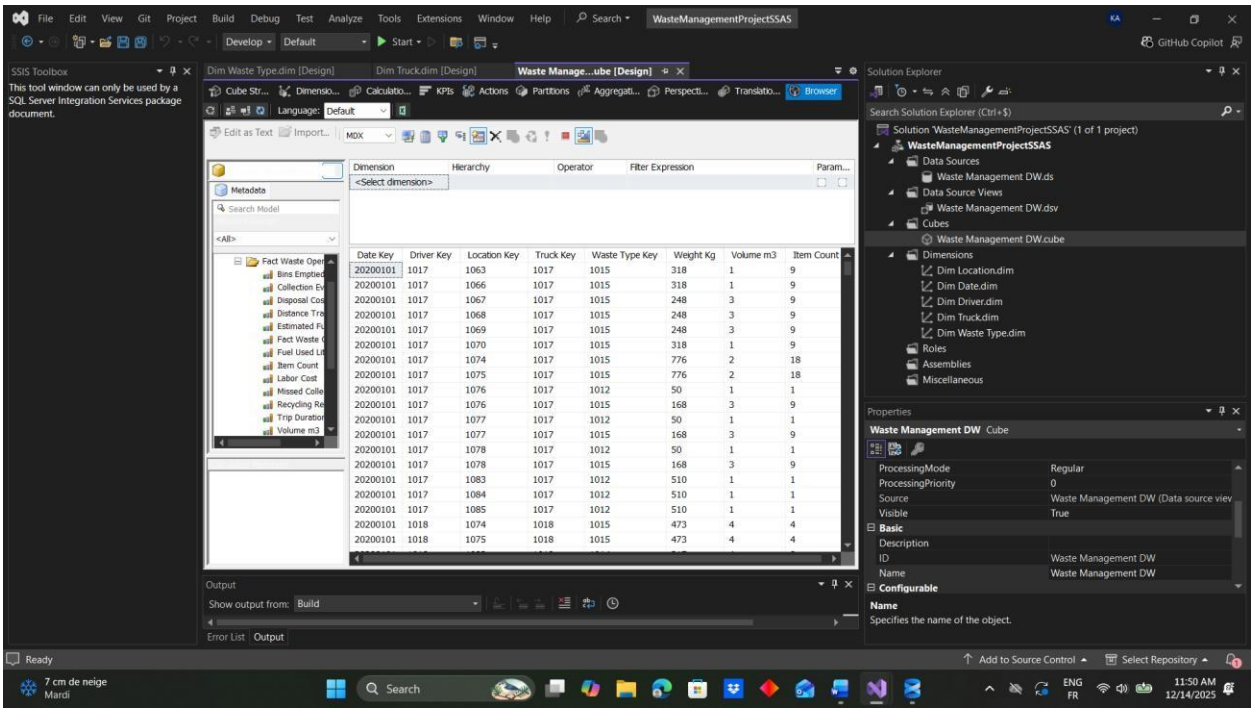


Figure 15: Data Successfully Added into the Cube

7.2 Measure Groups and Measures Definition:

The main measure group includes:

- Total waste collected
- Fuel consumption
- Trip duration
- Cost measures
- Recycling indicators

And we have added a “Total Trip Cost” measure to get the total cost of each trip

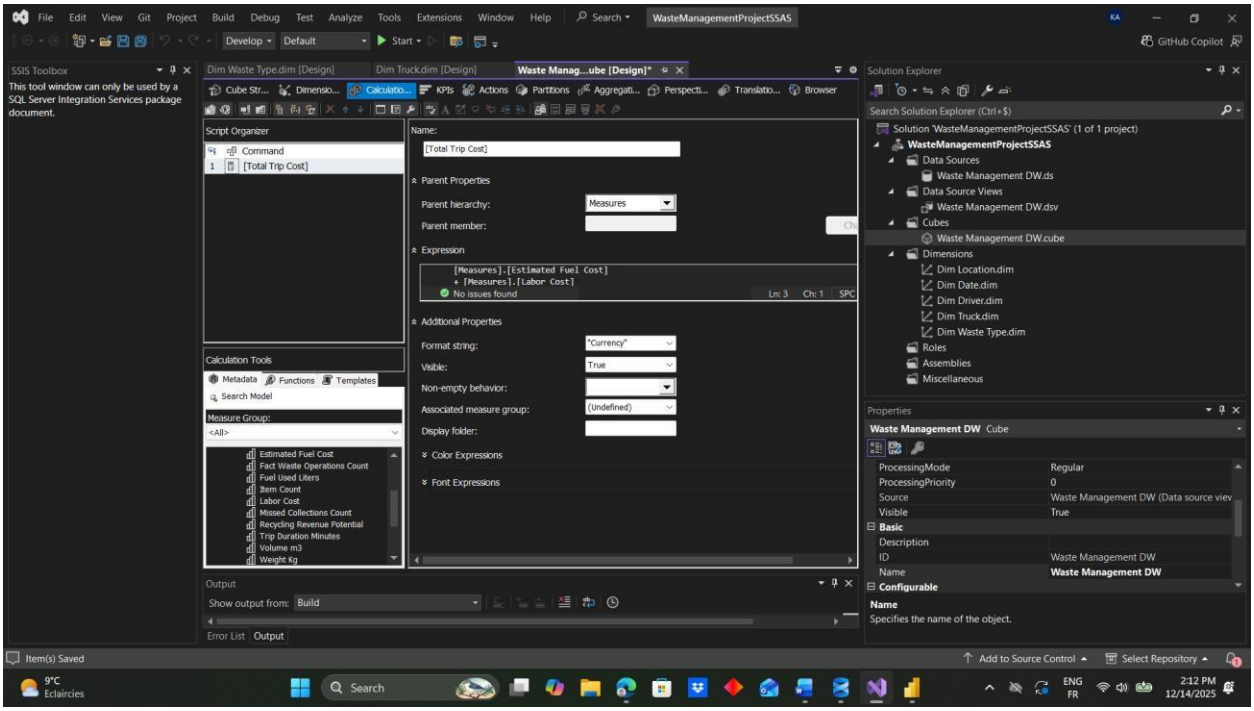


Figure 16: Total Trip Cost Measure

We have checked the added “Total Trip Cost” measure by displaying each truck code (e.g., TRK01–TRK10) associated with its corresponding trip count and total collected weight.

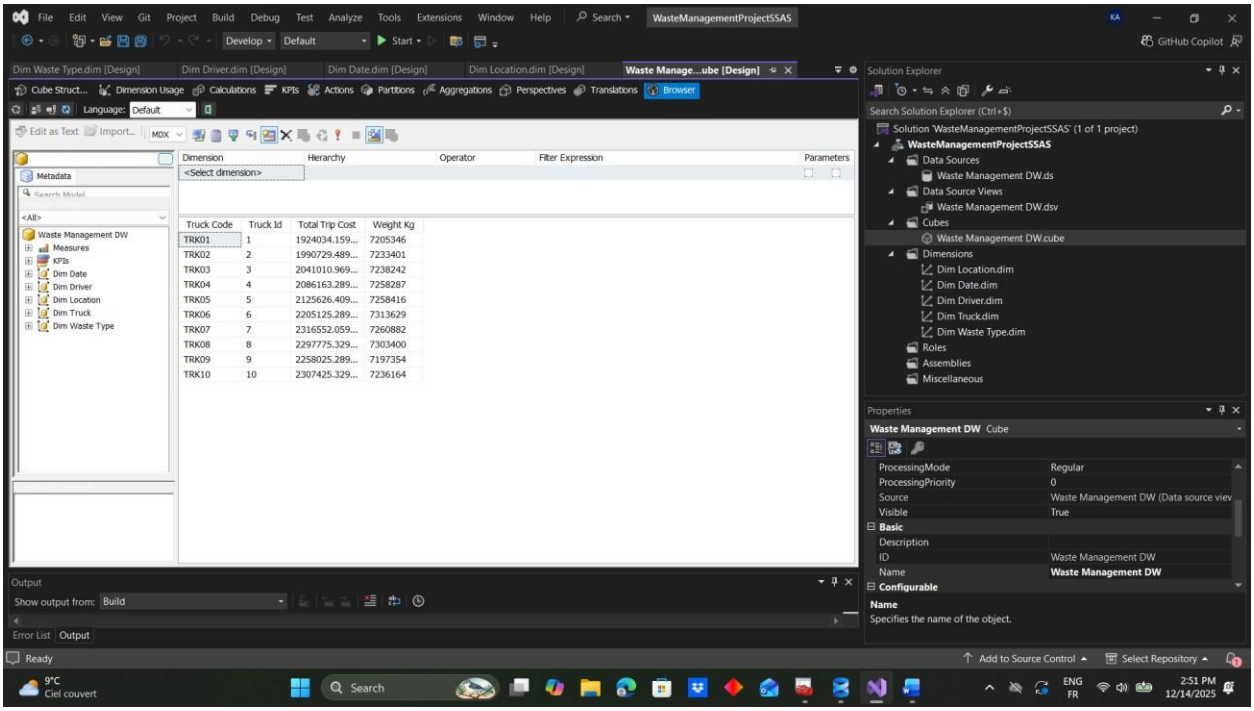


Figure 17: SSAS Cube Browser Displaying Waste Management Measures by Truck Dimension

7.3 Dimension Hierarchies and Relationships:

Hierarchies include:

- Date → Year → Month → Day
- Location → City → Zone
- Driver → Individual Driver
- Waste Type → Recyclable / Non-Recyclable

7.4 Calculated Measures and KPIs:

Calculated measures include:

- Collection efficiency (kg/hour)
- Fuel efficiency (km/liter)
- Recycling rate
- Cost per ton collected

7.5 Sample MDX Queries and Analysis:

This section presents selected MDX queries executed on the Waste Management DW cube using SQL Server Analysis Services (SSAS). These queries demonstrate the analytical capabilities of the multidimensional model and illustrate how operational data is transformed into actionable insights.

7.5.1 Waste Collection Performance by City (Top Waste Generators):

The following MDX query identifies the top five cities generating the highest total amount of waste, measured in kilograms. The TOPCOUNT function is used to rank cities based on the aggregated Weight Kg measure.

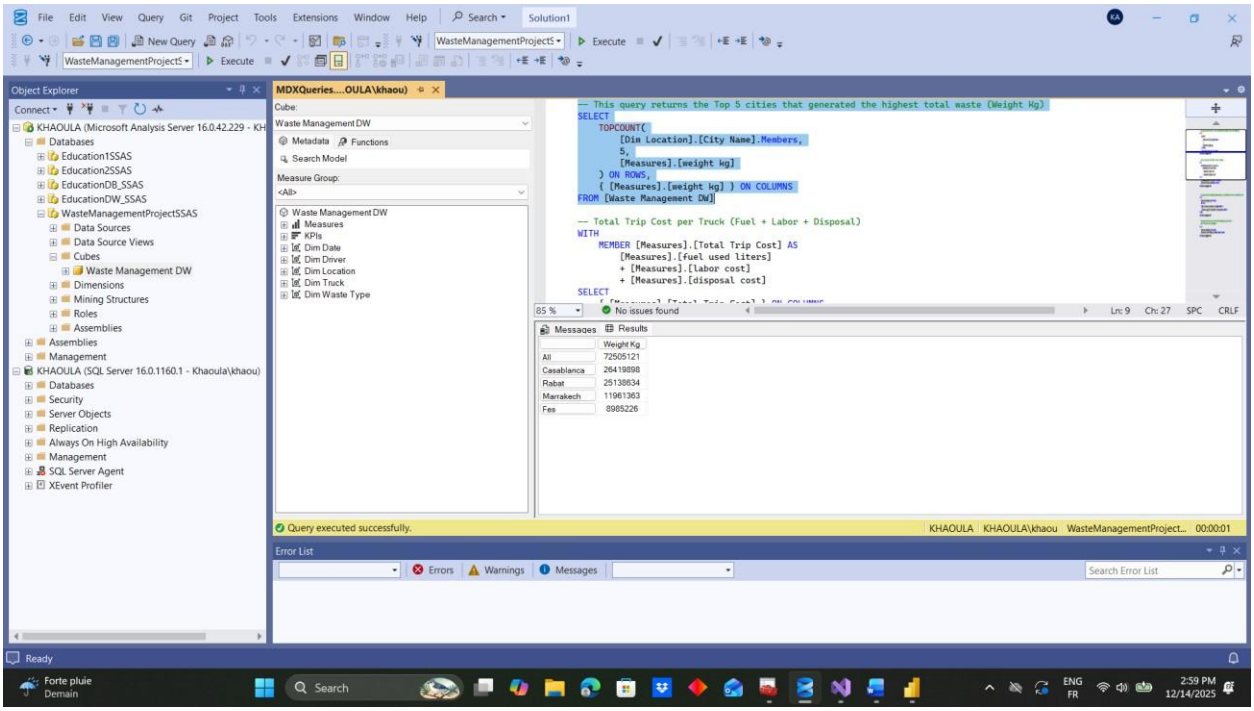


Figure 18: Top Waste Generators

Analysis

This query enables municipal decision-makers to quickly identify the cities contributing the most to total waste generation. Such insight is crucial for prioritizing resource allocation, infrastructure investment, and targeted waste reduction initiatives. High-performing cities may require additional collection capacity or recycling programs.

7.5.2 Total Trip Cost Analysis per Truck

To evaluate the operational cost efficiency of the vehicle fleet, a calculated measure is defined that aggregates fuel cost, labor cost, and disposal cost into a single metric named *Total Trip Cost*. This calculated measure is then analyzed per truck.

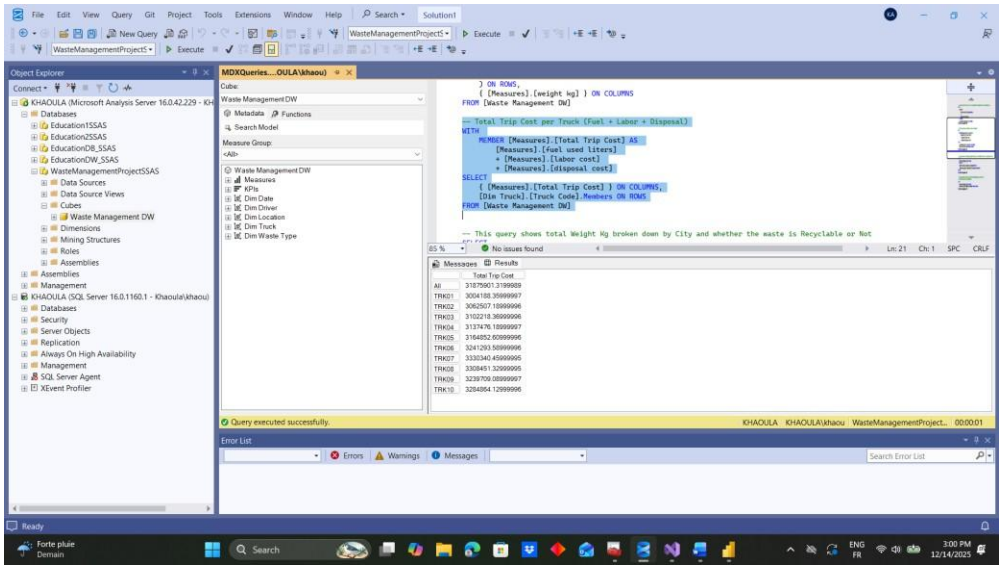


Figure 19: Total Trip Cost Analysis per Truck

Analysis

This query provides a clear comparison of total operational cost per truck. It helps identify high-cost vehicles, which may indicate inefficient routing, excessive fuel consumption, or aging fleet assets. These insights support maintenance planning and fleet optimization strategies.

7.5.3 Waste Composition Analysis by City and Recyclability

The following MDX query analyzes waste composition across cities, distinguishing between recyclable and non-recyclable waste. A cross-join between city and recyclability attributes is used to break down waste totals.

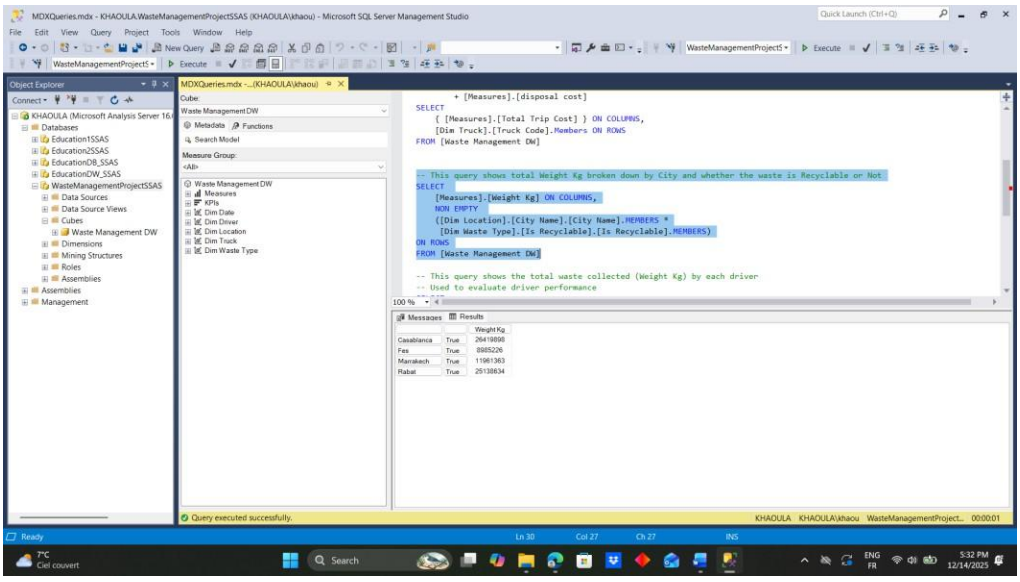


Figure 20: Waste Composition Analysis by City and Recyclability

Analysis

This query supports environmental sustainability analysis by highlighting how much recyclable waste is generated per city. It allows authorities to assess recycling effectiveness and identify cities where recyclable materials are underutilized, thus revealing opportunities for increased recycling revenue and environmental impact reduction.

7.5.4 Driver Performance Analysis

To evaluate workforce performance, the following MDX query computes the total amount of waste collected by each driver. The results provide a direct comparison of individual productivity.

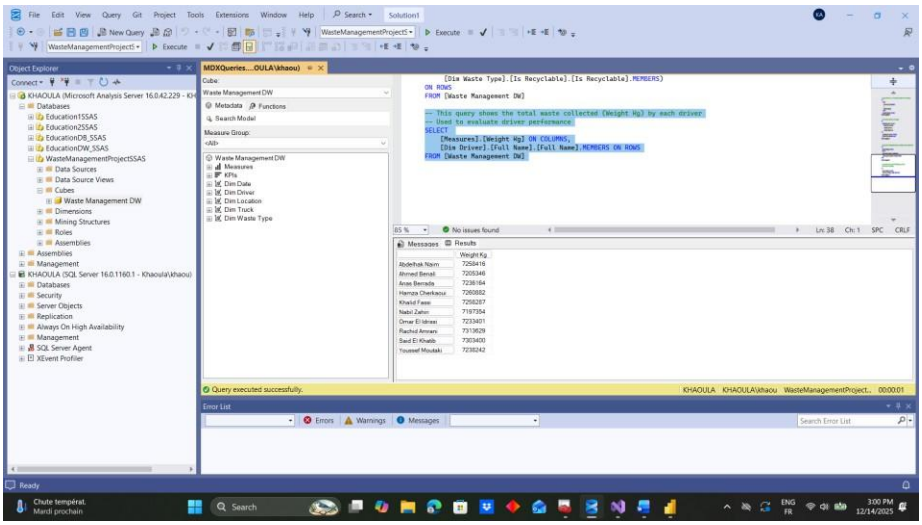


Figure 21: Driver Performance Analysis

Analysis

This query enables management to assess driver productivity based on total collected waste. It can be used to identify top-performing drivers, detect potential inefficiencies, and support performance-based incentives or targeted training programs.

7.5.5 Summary of MDX-Based Insights

The executed MDX queries demonstrate how the multidimensional cube supports:

- Strategic decision-making through ranked and aggregated views
- Cost control via calculated measures
- Sustainability assessment through waste composition analysis
- Operational monitoring through driver and vehicle performance evaluation

Together, these queries validate the effectiveness of the SSAS cube as a powerful analytical engine for municipal waste management.

8. REPORTING AND DATA VISUALIZATION

8.1 Reporting Architecture Overview

The reporting layer represents the final stage of the waste management decision-support system, transforming processed analytical data into interactive, user-friendly, and actionable insights. This layer is designed to serve multiple stakeholders, including municipal decision-makers, operational supervisors, and strategic planners.

The reporting architecture is built on a hybrid reporting approach, combining SQL Server Reporting Services (SSRS) for structured, parameter-driven reports and Power BI for interactive dashboards and exploratory analysis. Both reporting tools consume data from the centralized WasteManagementDW data warehouse and, where applicable, from the SSAS multidimensional cube.

Reporting Workflow

1. Operational data is collected in the OLTP system
2. Data is transformed and loaded into the Data Warehouse via SSIS
3. Analytical models are exposed through SSAS and SQL views
4. SSRS and Power BI consume curated datasets
5. Users interact with reports through parameters, charts, and dashboards

8.2 SQL Server Reporting Services (SSRS)

8.2.1 Executive Dashboard Report

Provides high-level KPIs for decision-makers.

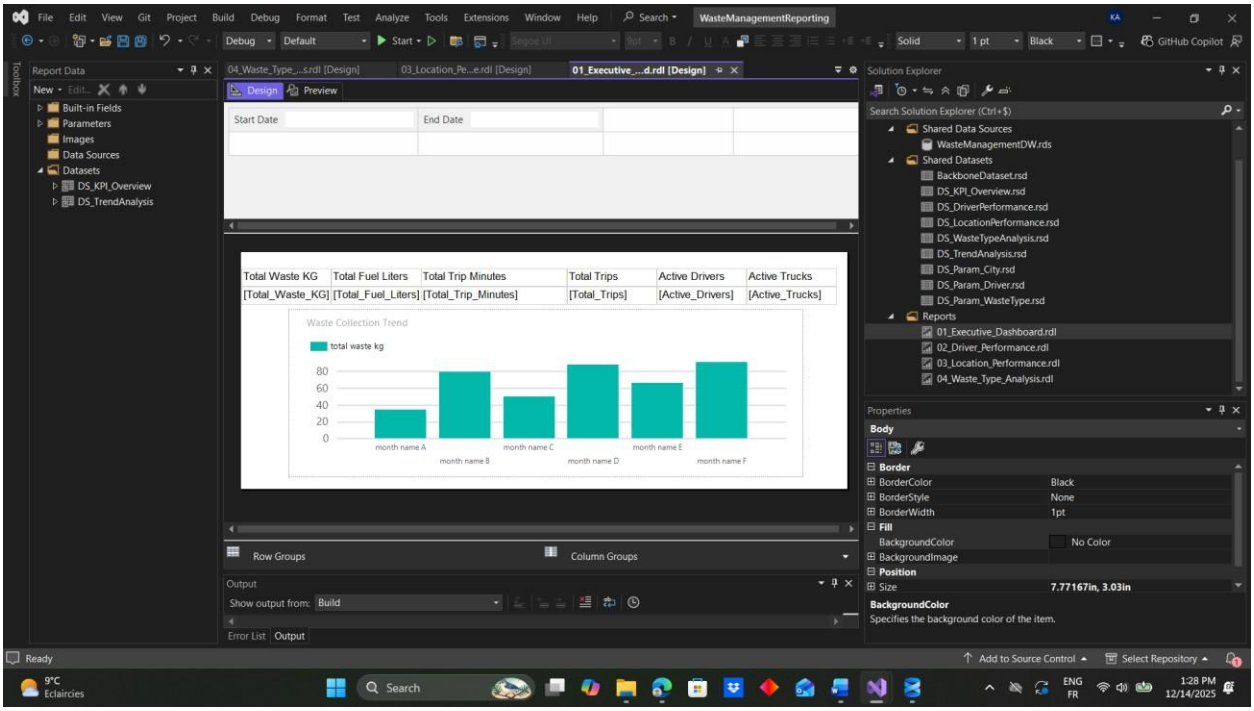


Figure 22: Executive SSRS Dashboard Desigfor Waste Collection KPIs and Monthly Trend Analysis

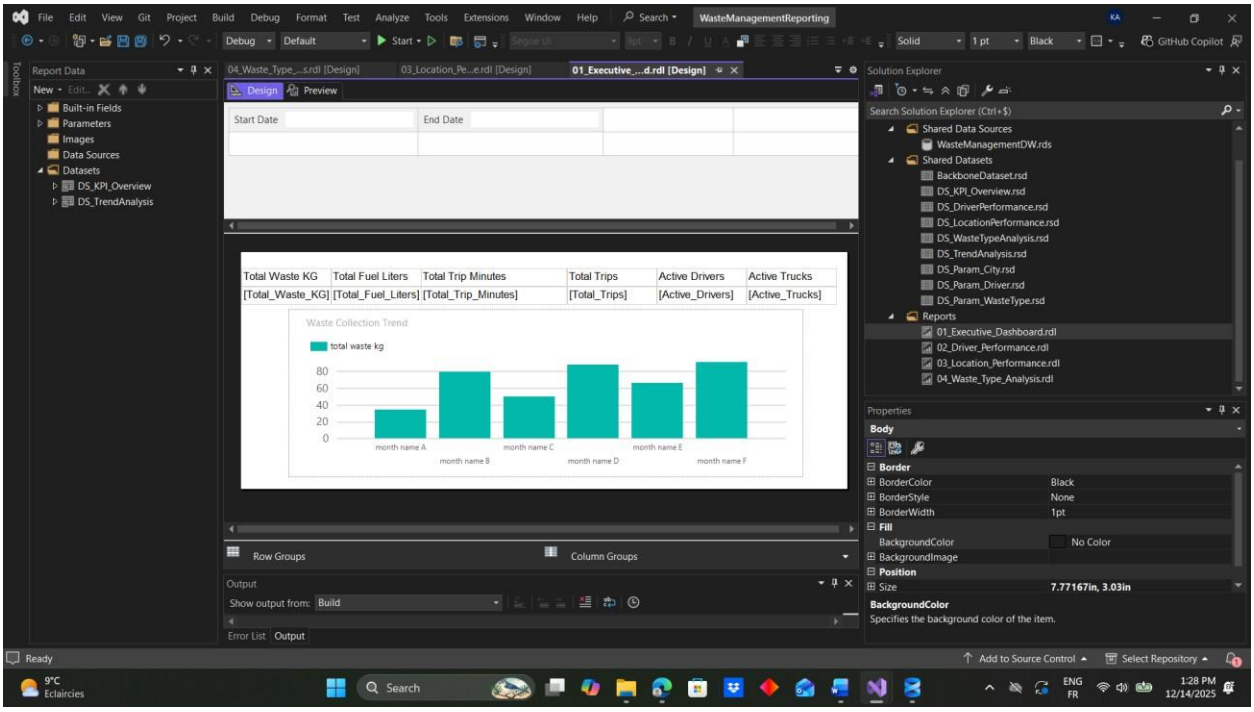


Figure 23: Executive SSRS Dashboard Report Displaying Waste Collection KPIs and Monthly Trend Analysis

8.2.2 Driver Performance Report

Analyzes productivity and efficiency per driver.

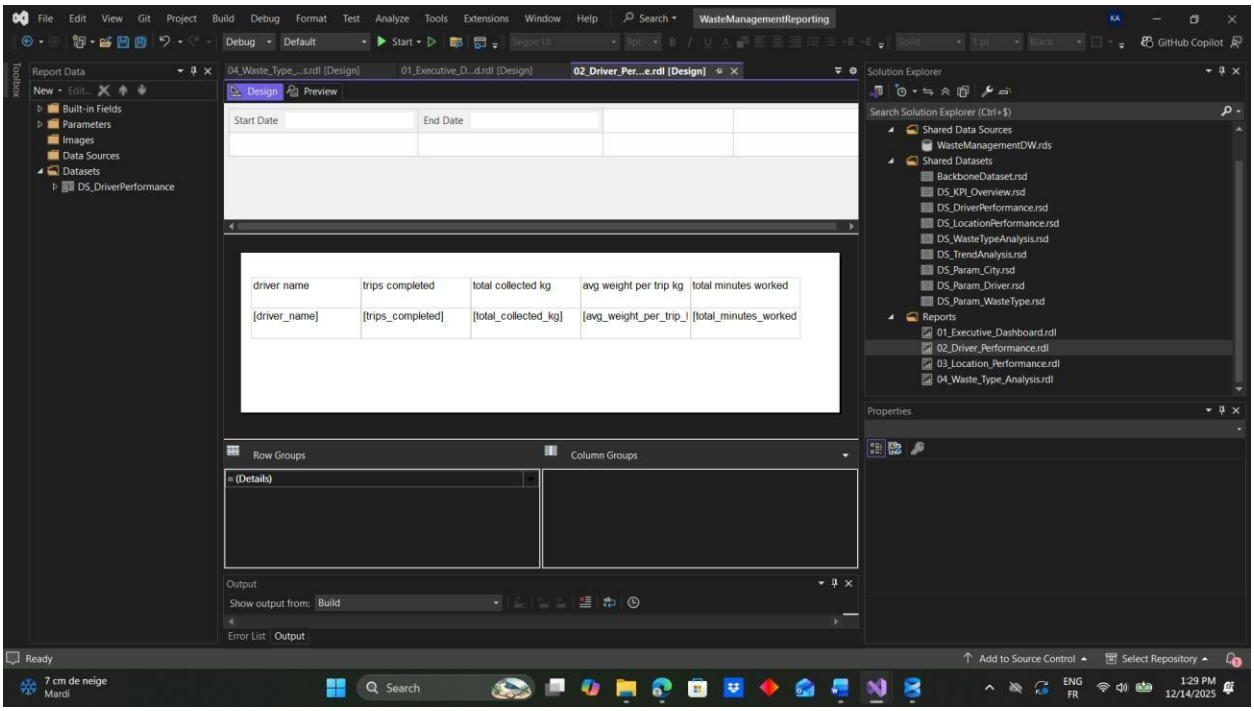


Figure 24: Driver Performance SSRS Report Design Showing Key Operational Metrics

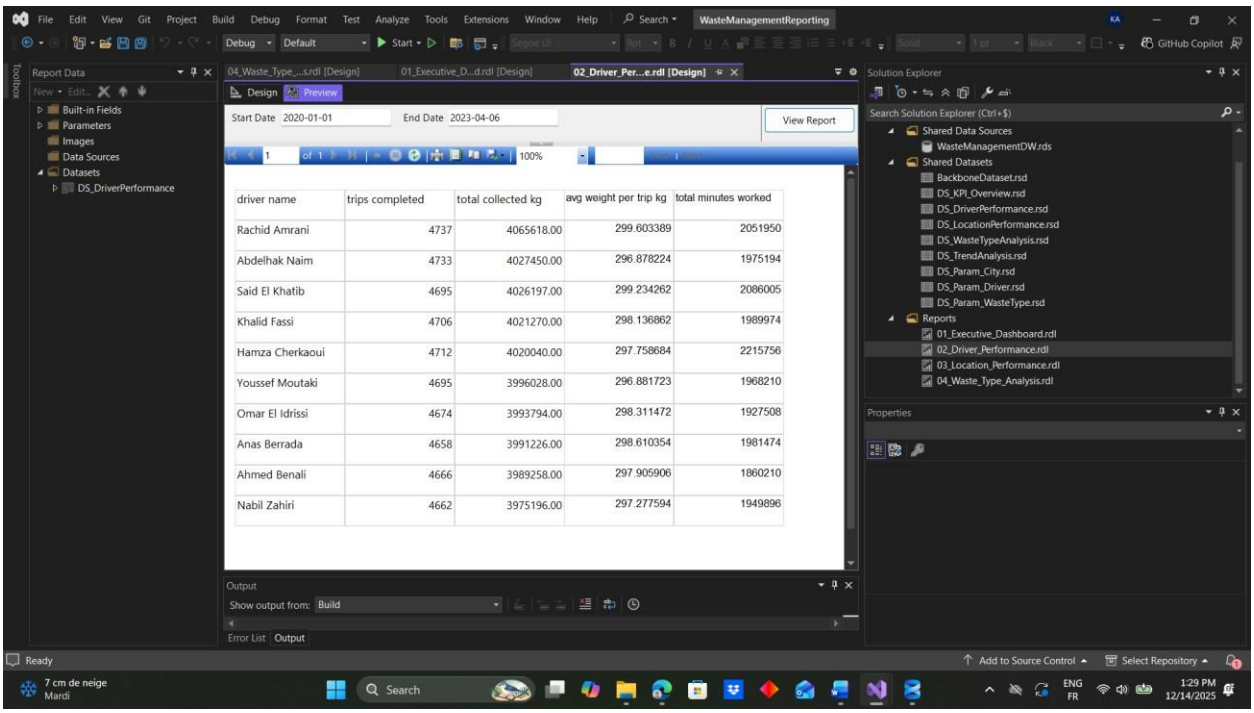


Figure 25: Driver Performance Report Preview Displaying Trips, Collected Weight, and Working Time

8.2.3 Location Performance Report

Highlights high- and low-performing zones.

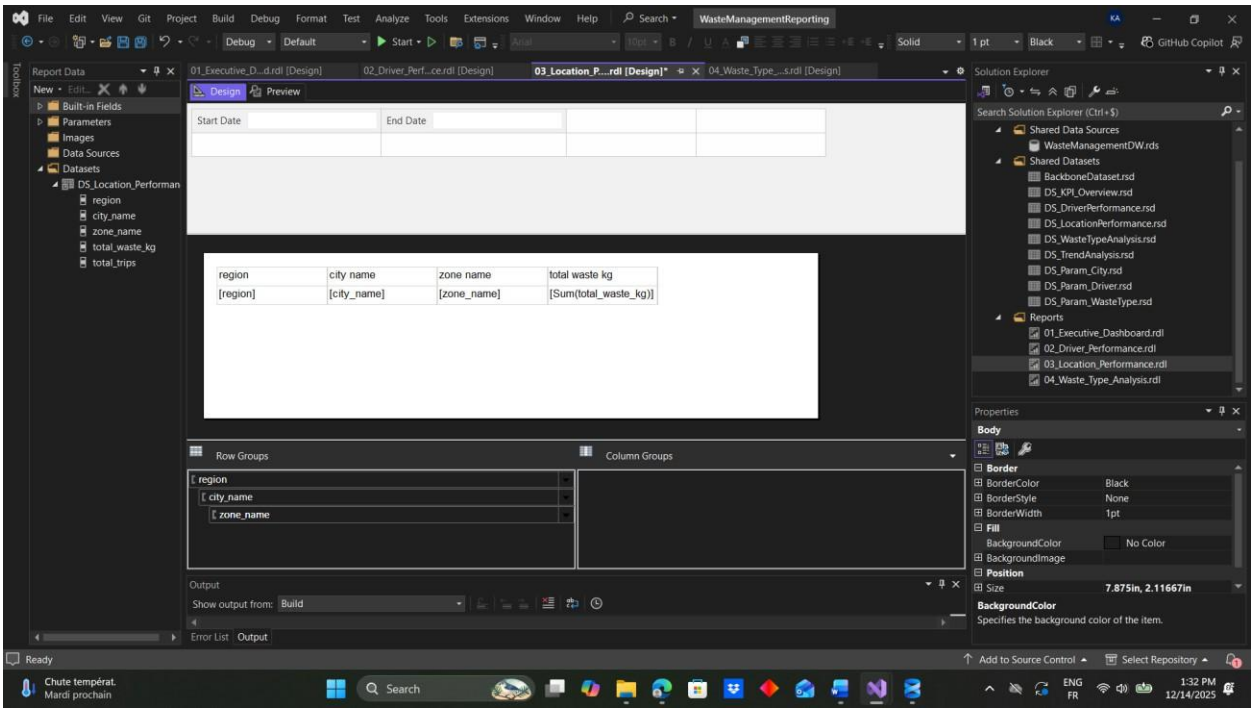
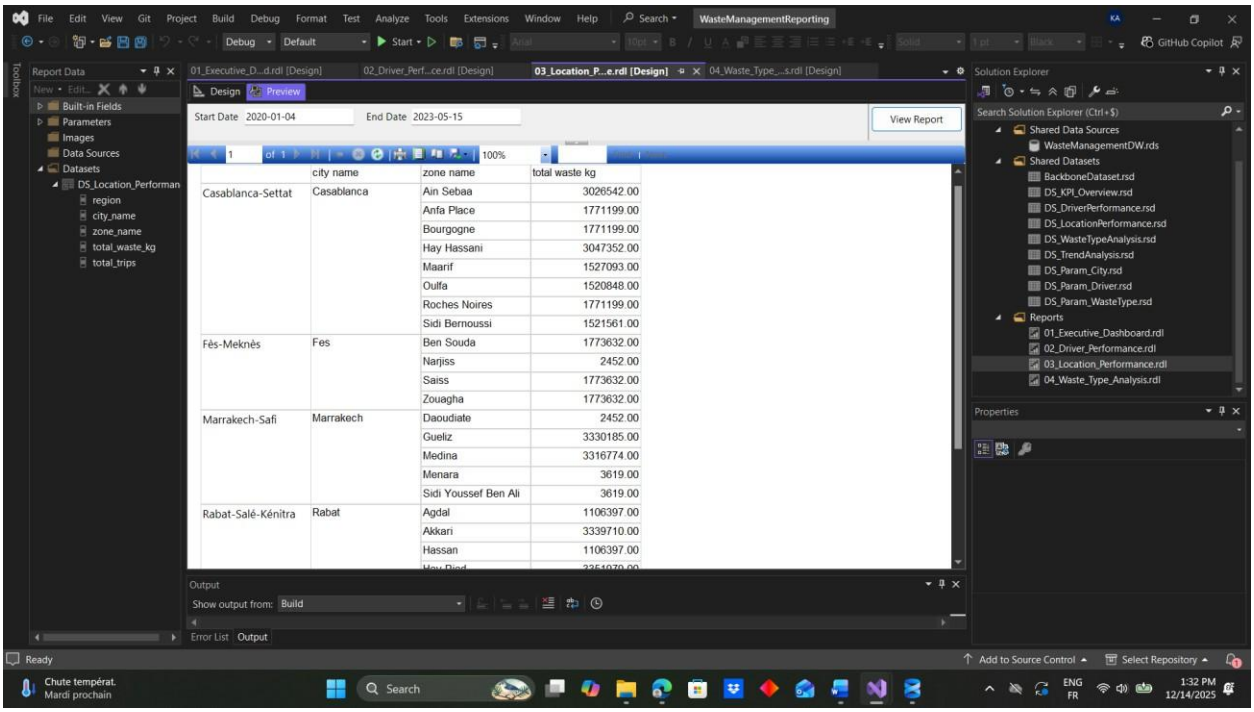


Figure 26: Location Performance SSRS Report Design with Hierarchical Grouping by Region, City, and Zone



city name	zone name	total waste kg
Casablanca-Settat	Casablanca	3026542.00
	Ain Sebaa	1771199.00
	Anfa Place	1771199.00
	Bourgogne	3047352.00
	Hay Hassani	1527093.00
	Maarif	1520848.00
	Oulfa	1771199.00
	Roches Noires	1521561.00
	Sidi Bernoussi	1773632.00
Fès-Meknès	Fes	2452.00
	Ben Souda	1773632.00
	Narjiss	1773632.00
	Saiss	1773632.00
	Zouagha	2452.00
Marrakech-Safi	Marrakech	3330185.00
	Daoudiate	3316774.00
	Gueliz	3619.00
	Menara	3619.00
	Medina	1106397.00
	Sidi Youssef Ben Ali	3339710.00
Rabat-Salé-Kénitra	Rabat	1106397.00
	Agdal	
	Akkari	
	Hassan	

Figure 27: Location Performance Report Showing Total Waste Collected by Region, City, and Zone

8.2.4 Waste Type Analysis Report

Displays waste composition and recycling potential.

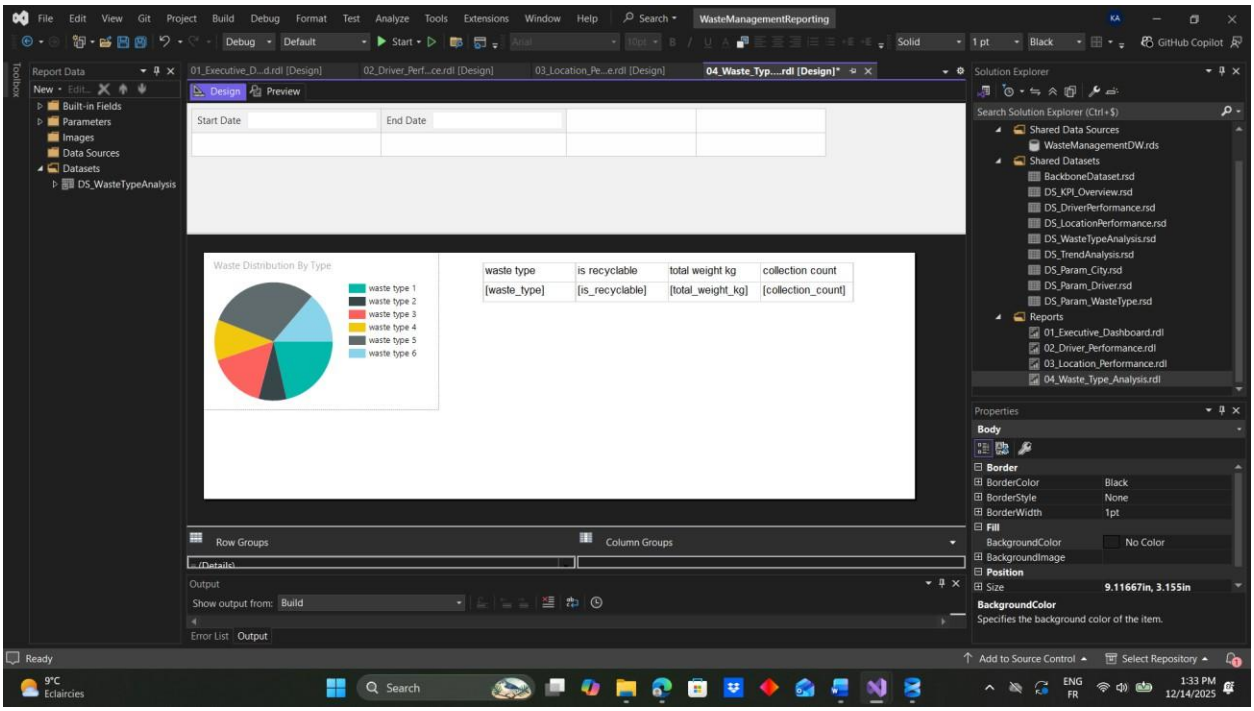


Figure 28: Waste Type Analysis SSRS Report Design with Distribution Chart and Summary Metrics

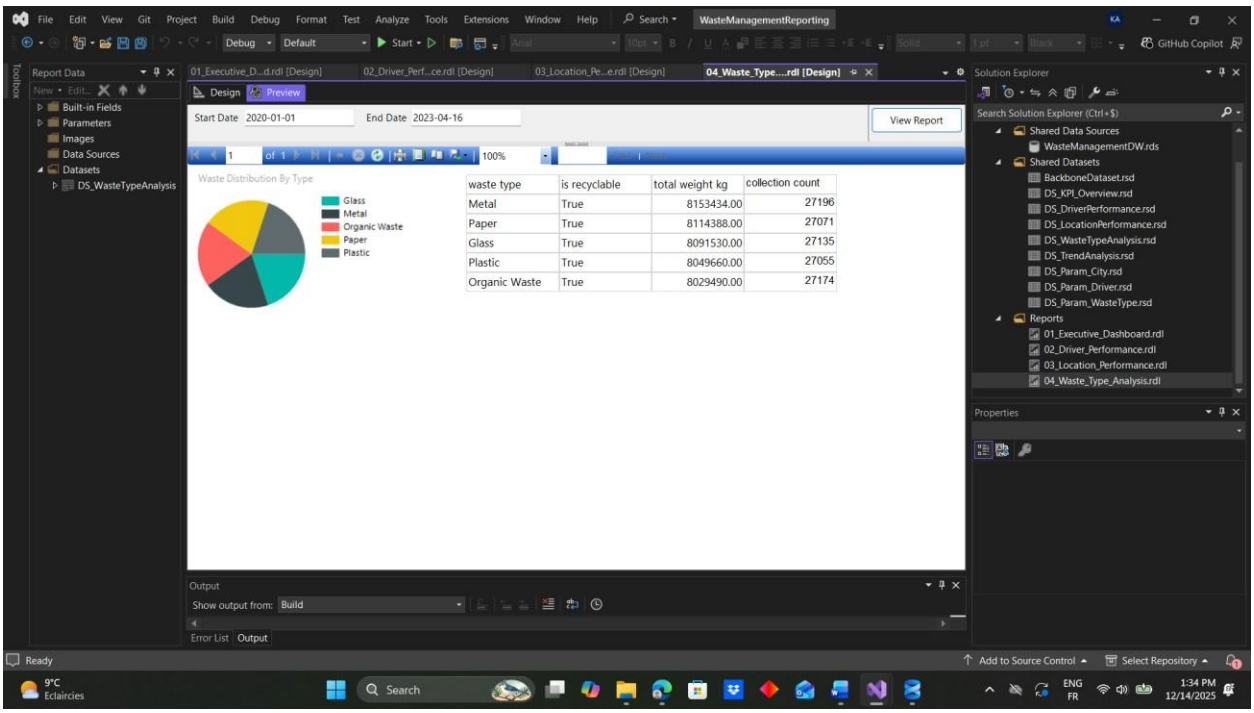


Figure 29: Waste Type Analysis Report Showing Distribution, Recyclability, Total Weight, and Collection Count

8.3 Power BI Dashboard

The Power BI dashboard provides an interactive analytical environment that complements the static and parameter-driven SSRS reports. It enables decision-makers to explore waste management data dynamically, identify trends, compare performance across cities, and assess operational costs in real time. The dashboard is built on a live connection to the analytical model and presents multiple coordinated visualizations.

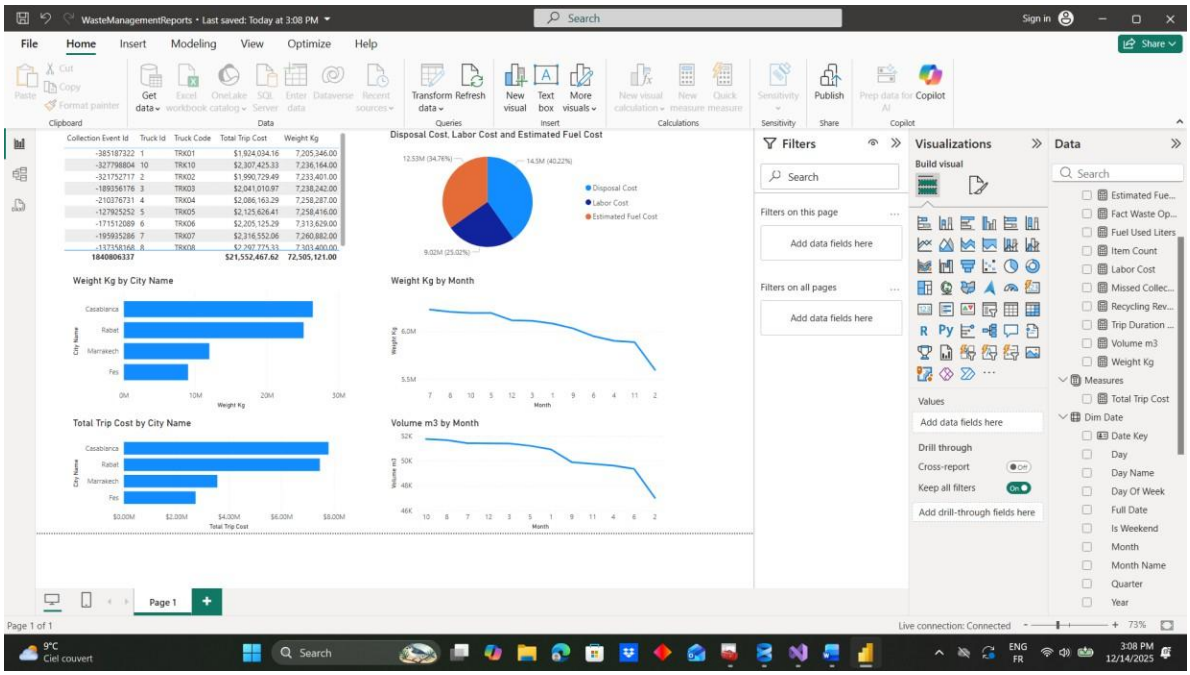


Figure 30: Power BI Waste Management Dashboard

8.3.1 Waste Collection and Cost Overview Table

Description

The tabular visual at the top of the dashboard presents a detailed breakdown of waste collection operations at the transaction level. It includes the following key attributes:

- Collection Event ID
- Truck ID and Truck Code
- Total Trip Cost
- Weight Collected (Kg)
- Disposal Cost
- Labor Cost
- Estimated Fuel Cost

Purpose
This table allows users to inspect individual collection events and understand how operational costs accumulate per trip. It serves as a validation and drill-down tool supporting the aggregated visualizations.

Insight
High-cost trips can be quickly identified and traced back to specific trucks or collection events, enabling targeted operational audits.

8.3.2 Cost Distribution Analysis (Pie Chart)

Description
The pie chart displays the proportional distribution of:

- Disposal Cost
- Labor Cost
- Estimated Fuel Cost

across total waste management operations.

Purpose
This visualization highlights the relative contribution of each cost component to the overall operational expenditure.

Insight
The chart reveals which cost category dominates total expenses, helping decision-makers prioritize cost optimization strategies, such as fuel efficiency programs or labor scheduling improvements.

8.3.3 Waste Collected by City (Bar Chart)

Description
The horizontal bar chart titled “Weight Kg by City Name” compares the total weight of waste collected across major cities, including Casablanca, Rabat, Marrakech, and Fes.

Purpose
This visual supports spatial comparison of waste generation and collection volume across urban areas.

Insight

Cities with significantly higher waste volumes can be identified as priority zones for infrastructure investment, route optimization, or recycling initiatives.

8.3.4 Total Trip Cost by City (Bar Chart)

Description

This bar chart presents the total trip cost aggregated by city.

Purpose

It allows users to compare operational expenses geographically and evaluate cost efficiency at the city level.

Insight

Discrepancies between waste volume and cost may indicate inefficiencies, such as longer routes, higher fuel consumption, or operational constraints in certain cities.

8.3.5 Monthly Waste Collection Trend (Line Chart)

Description

The line chart titled “Weight Kg by Month” illustrates the evolution of total waste collected over time.

Purpose

This visualization enables trend analysis and seasonal pattern detection in waste generation.

Insight

Downward or upward trends can indicate behavioral changes, policy impacts, or seasonal effects, supporting data-driven planning and forecasting.

8.3.6 Monthly Volume Trend (Line Chart)

Description

The “Volume m3 by Month” line chart tracks changes in the volume of waste collected over time.

Purpose

It complements the weight-based analysis by highlighting volumetric capacity requirements.

Insight

Differences between weight and volume trends may reveal changes in waste composition,

such as increased recyclable or lightweight materials, which is critical for fleet capacity planning.

8.3.7 Dashboard Interactivity and Filtering

Description

All visuals are interconnected through Power BI's cross-filtering functionality. Selecting a city, month, or cost component dynamically updates all other visuals.

Purpose

This interactivity enables exploratory analysis without requiring predefined queries.

Insight

Users can quickly isolate problem areas, compare scenarios, and gain deeper operational understanding through intuitive interactions.

9. RESULTS AND DISCUSSION

9.1 Operational Performance & Efficiency Analysis

The analytical results derived from the data warehouse and multidimensional cube highlight significant insights into the operational performance of waste collection activities across major Moroccan cities. By aggregating waste collection metrics such as total weight collected, trip duration, number of trips, and distance traveled, the system provides a comprehensive view of operational efficiency.

Trend analysis reveals variations in waste generation and collection intensity across time and geographic locations, indicating opportunities for operational optimization. Cities with higher waste volumes and longer trip durations were identified, allowing decision-makers to evaluate route planning effectiveness and resource allocation. The availability of performance indicators at multiple levels of granularity (city, zone, driver, and truck) enables targeted operational improvements rather than generalized interventions.

Overall, the system demonstrates its capacity to transform raw operational data into structured performance metrics, supporting evidence-based optimization of waste collection operations.

9.2 Cost and Fuel Consumption Analysis

Cost-related measures, including fuel consumption, labor costs, disposal costs, and total trip cost, were analyzed using both SSAS calculated measures and reporting tools. The integration of these metrics allows for a holistic assessment of operational expenses associated with waste collection activities.

The analysis highlights substantial variations in fuel usage and trip costs across different trucks, drivers, and cities. These discrepancies suggest inefficiencies related to route length, vehicle utilization, and operational practices. By correlating distance traveled with fuel consumption, the system enables the identification of high-cost routes and inefficient vehicle deployments.

The cost analysis demonstrates the value of centralized analytical reporting in supporting financial oversight and cost-control strategies. Municipal authorities and service providers can leverage these insights to refine routing strategies, improve fleet utilization, and reduce unnecessary expenditures.

9.3 Recycling Performance Analysis

Recycling performance was evaluated by analyzing waste collection volumes based on recyclability attributes. The results indicate an imbalance between recyclable and non-recyclable waste streams across cities, highlighting underutilized recycling potential.

The system enables the segregation of waste volumes by recyclability status and location, allowing stakeholders to identify areas with high recyclable waste generation that are not adequately captured by existing recycling processes. These insights can support strategic investments in recycling infrastructure and awareness programs.

Although the system does not perform predictive modeling, it provides a reliable analytical foundation for estimating recycling revenue potential based on historical waste composition trends, contributing to data-driven environmental and economic planning.

9.4 Service Monitoring and Analytical Insights

Service performance monitoring was achieved through the analysis of trip completion, collection frequency, and operational coverage across time and geographic dimensions. While the system does not operate in real time, periodic reporting provides valuable insights into service consistency and operational reliability.

Temporal trend analysis highlights fluctuations in service intensity and waste collection volumes, enabling planners to assess seasonal and monthly variations. Spatial analysis at the city and zone levels supports performance benchmarking and identification of service gaps.

The analytical framework thus enhances transparency and accountability by providing measurable indicators of service delivery, which are essential for continuous improvement and performance evaluation.

9.5 Impact of the Analytical Platform on Decision Support

The integration of SQL Server Analysis Services, SQL Server Reporting Services, and Power BI establishes a unified analytical platform that significantly enhances decision-support capabilities. Executives and operational managers can access consistent, validated metrics through interactive dashboards and structured reports.

The platform supports both strategic and operational decision-making by enabling drill-down analysis, trend evaluation, and comparative performance assessments. The separation between data storage, analytics, and visualization layers ensures scalability, data governance, and analytical consistency.

Overall, the implemented solution demonstrates how modern business intelligence architectures can improve decision quality, operational awareness, and strategic planning in municipal waste management contexts.

10. CONCLUSION AND FUTURE WORK

10.1 Project Summary & Contributions

This project successfully designed and implemented an end-to-end business intelligence solution for municipal waste management using Microsoft SQL Server technologies. The solution integrates an OLTP database, a star-schema data warehouse, an SSIS-based ETL pipeline, an SSAS multidimensional cube, and comprehensive reporting layers using SSRS and Power BI.

The system consolidates fragmented operational data into a centralized analytical environment, enabling structured analysis of waste collection performance, costs, and recycling potential across major Moroccan cities.

10.2 Key Achievements

The implemented platform delivered several key analytical outcomes:

- Centralized and cleansed operational data suitable for analytical processing
- Multidimensional analysis of waste collection metrics across time, location, and operational entities
- Cost and fuel consumption insights supporting financial optimization
- Interactive reports and dashboards enabling intuitive exploration of performance indicators

These outcomes demonstrate the effectiveness of the architecture in supporting data-driven waste management strategies.

10.3 Limitations of the Current System

- The system is based on daily data refreshes, which means it does not provide real-time monitoring of waste collection activities.
- The analysis focuses mainly on historical and descriptive insights; predictive analytics and optimization features are not included.
- Cost calculations (fuel, labor, disposal) are based on estimated business rules, which may differ from real-world fluctuations.
- The solution relies only on internal operational data, without integration of external sources such as traffic, weather, or population data.
- Advanced data governance features, such as automated anomaly detection or data quality alerts, are not yet implemented.

10.4 Recommendations and Future Enhancements

- Introduce real-time data collection using GPS trackers, IoT sensors, or mobile applications for trucks and drivers.
- Implement predictive analytics models to forecast waste generation and optimize routes proactively.

- Enhance cost models by integrating actual fuel prices, labor contracts, and disposal fees.
- Integrate external data sources (traffic, weather, population growth) to improve operational planning accuracy.
- Strengthen data governance and security by adding automated data quality checks and audit mechanisms.

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